MIL-T-35230(AS)
27 March 1984
SUPERSEDING
SD-4320
8 July 1980
AS-4320
20 Sept 1975

MILITARY SPECIFICATION

TARGET, TOW, AERIAL, TDU-34/A

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the design, fabrication and inspection of the TDU-34/A Aerial Tow Target herein referred to as the target.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636	Box, Shipping, Fiberboard

MILITARY

MIL-C-6021	Castings, Classification And Inspection Of
MIL-A-8591	Airborne Stores, Associated Suspension Lugs And Aircraft - Store Interface (Carriage Phase), General Design Criteria For
MIL-N-18307	Nomenclature And Identification For Electronic Aeronautical, And Aeronautical Support Equipment Including Ground Support Equipment

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to the Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD), Code 93, Lakehurst, NJ 08733 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1550

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables For Inspection By Attributes
MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of U.S. Military Property
MIL-STD-210	Climatic Extremes For Military Equipment
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-794	Parts And Equipment, Procedures For Packaging And Packing Of
MIL-STD-882	System Safety Program Requirements
MIL-STD-965	Parts Control Program
MIL-STD-1521	Technical Reviews And Audits For Systems, Equipment And Computer Programs

2.1.2 Other Government documents, drawings, and publications. The following other Government documents and drawings form a part of this specification to the extent specified herein.

DRAWINGS

NAVY

NAVAL AIR DEVELOPMENT CENTER, CODE IDENT NO. 80260

TE21053	Target Adapter Assembly

PDA ENGINEERING, CODE IDENT 55794

573011	Packaging Assembly
573238	Radar Reflective Nose Cone
573239	Aft Reflector Assembly
573849	Body Assembly
573850	TDU-34/A Assembly
573856	Fin
573858	Radar Non-reflective Nose Cone
573948	TAS Rack
573962	CG Fixture

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Item description. The target shall be an aerodynamic stabilized airframe capable of being air launched, streamed, towed, and air recovered from fighter and attack aircraft equipped with either an A/A47U-3 or A/A47U-4 Tow Target Reeling Machine Launcher System.

3.1.1 Major component list. The target shall consist of the following major components.

a. Body Assembly, (see Drawing 573849)

b. Radar Reflective Nose Cone, (see Drawing 573238)

c. Aft Reflector Assembly, (see Drawing 573239)

d. Fins (4), (see Drawing 573856)

e. Target Adapter Assembly, (see Drawing TE21053)

3.1.2 Government-loaned property. The target shall include complete provisions (see 6.3.1) for the following items of Government-loaned property (see 6.2.1f).

a. Radar Non-reflective Nose Cone, (see Drawing 573858)

b. TAS Rack, (see Drawing 573948)

3.1.3 First article sample. When specified, a sample shall be subjected to first article inspection (see 4.2.1 and 6.2.1g).

3.2 Characteristics.

3.2.1 Performance characteristics. The performance characteristics shall apply after transportation, unpackaging, and reassembly of the target which has been fabricated and assembled in accordance with the drawings (see 3.3.1), inspected and balanced in accordance with Section 4 herein, and packaged in accordance with Section 5 herein. Engineering change proposals to the drawings shall not degrade the following performance characteristics of the target (see 6.2.1e).

3.2.1.1 Flight. The target after installation on the aircraft shall be capable of 10 successive stowed and maneuvering towed flights of two hours duration each within the flight envelope of Figure 1.

3.2.1.2 Minimum radar cross section (RCS). When irradiated by any monostatic, linearly polarized radar operating at 10 Gigahertz (GHz), the target shall exhibit the RCS signature of Appendix A.

3.2.2 Physical characteristics. Engineering change proposals to the drawings shall not degrade the following physical characteristics of the target (see 6.2.1e).

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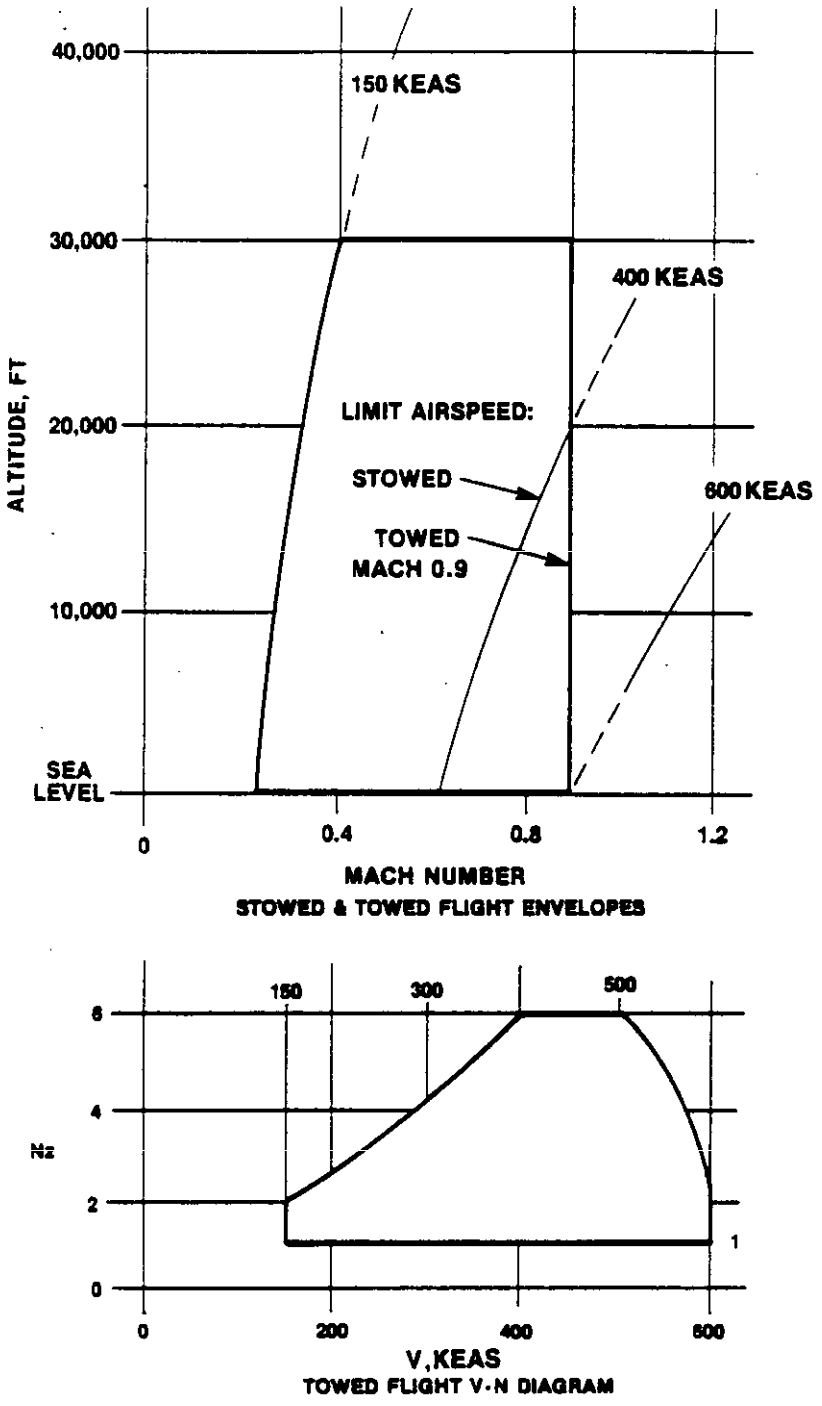


FIGURE 1. Flight and load envelopes

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3.2.2.1 Weight and center of gravity. The weight of the target (without Government loaned property) shall be not greater than 80 pounds. The center of gravity shall be 2.0 plus or minus 0.5 inches aft of the target tow point.

3.2.2.2 Structural Strength. The target structure shall survive design limit loads imposed by the catapult, arrestment and flight of the carrying aircraft, which shall be computed based on MIL-A-8591 using the loads computation method specified in Appendix B. The target shall repetitively withstand without failure the load resulting from a target to launcher closure rate of not greater than two and one half feet per second. When resting on the nose and two fins, the target shall withstand the loads associated with a vertical down limit load factor of not less than three.

3.2.2.3 Survivability. The target in towed flight shall be capable of absorbing 20 millimeter projectile hits without propagation of damage except where structural stiffeners (i.e., forward launcher support casting, aft launcher and tow strongback casting, and fin mounting bulkheads) are damaged by a projectile hit.

3.2.3 Environment. Engineering change proposals to the drawings shall not degrade the following environmental characteristics of the target (see 6.2.1e)

3.2.3.1 Operational environment. The targets shall be towable without failure when exposed to the 20 percent extremes of the Naval Air Environment of MIL-STD-210, and the aircraft shock and vibration environment (see 3.2.2.2).

3.2.3.2 Stowage environment. The target shall be towable without failure after exposure to the Naval Surface Environment of MIL-STD-210 when packaged (see 5.1) for a period of five years, and when stored unpackaged for three months.

3.3 Design and construction.

3.3.1 Drawings. The drawings shall be those listed on the issue of Drawing 573850 referenced in the contract (see 6.2.1b).

3.3.2 Standards of manufacture. Engineering change proposals to the drawings shall be subject to Government approval in accordance with the contract, and shall not degrade the following standards (see 6.2.1e).

3.3.2.1 Materials, processes, and parts. Materials for the nose and aft reflector assemblies shall be selected to provide radar reflectivity in accordance with Appendix A (see 3.2.1.2).

3.3.2.2 Standard items. Parts shall be in accordance with MIL-STD-965. Commercial utility parts such as screws, bolts, nuts, cotter pins, etc., may be used provided they are interchangeable (see 6.3.2) with AN or MIL standard parts, which shall be identified on the production drawing.

3.3.2.3 Interchangeability. Target parts having the same part number shall be interchangeable (see 6.3.2) between targets. Target fins shall be interchangeable between positions on each target. Assembly indices for the aft reflector assembly and the nose assembly shall be avoided.

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3.3.2.4 Manufacturing alignments. All alignments of fabricated and assembled target components specified on the drawings shall be controlled by tooling.

3.3.2.5 Castings. Castings shall be classified in accordance with MIL-C-6021.

3.3.2.6 Tow system interface. The target shall be physically and functionally compatible with either the RMK-19/A, or the RMK-31/A Reeling Machine Launchers of the A/A47U-3 or A/A47U-4 Tow Target Reeling Machine Launcher Systems.

3.3.2.7 Component assembly. Brackets, lugs, flanges, inserts, bolts, and other assembly provisions shall retain components securely during the life of the target. Parts shall be mounted and installed so loosening, wear of mounting arrangements, or permanent separation of parts shall not occur. Components shall be designed to be incapable of incorrect assembly.

3.4 Reliability. Not applicable to this specification.

3.5 Maintainability. Except for the target fuselage, the time to replace a major component shall be not greater than 10 minutes using standard Government tools.

3.6 Parts marking. Markings of parts shall be in accordance with MIL-STD-130 as indicated on the drawings.

3.7 Exterior color, markings and insignia. The target exterior color of International Orange, and target insignia and markings shall be in accordance with the drawings.

3.7.1 Nameplates. Nameplates shall be in accordance with MIL-N-18307.

3.8 Safety. A system safety program shall be established in accordance with MIL-STD-882, and maintained to ensure that proposed engineering changes shall not degrade design safety (see 6.2.1e).

3.9 Workmanship. Each target shall be uniform in quality and shall be free from irregularities or defects which could affect compliance with the requirements of this specification. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges. Assembly of parts shall be accomplished without damage to any part.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1)

b. Acceptance inspection (see 4.2.2)

c. Special inspections (see 4.2.3)

4.2.1 First article inspection. First article inspections will consist of tests, analyses and examinations by the Government to determine compliance with the specification requirements, and will be conducted whenever the contractor has not previously fabricated the target to the drawings, or when otherwise required by the Government (see 4.1). First articles shall have successfully passed the acceptance inspection at the contractor's facility prior to first article inspection by the Government (see 6.2.1g).

4.2.1.1 Configuration audit. The first article inspection shall include the acceptance inspection, and a physical configuration audit in accordance with the applicable requirements of Appendix F of MIL-STD-1521. The audit will assess compliance with the following requirements:

a. 3.1.1 Major components
b. 3.1.2 Government loaned property
c. 3.2.2 Physical characteristics
d. 3.2.2.1 Weight and C.G.
e. 3.3.1 Drawings
f. 3.3.2.1 Material, processes, parts
g. 3.3.2.2 Standard items
h. 3.3.2.3 Interchangeability
i. 3.3.2.4 Alignments
j. 3.3.2.7 Component Assembly
k. 3.5 Maintainability
l. 3.6 Parts Marking
m. 3.7 Exterior color, marking & insignia
n. 3.9 Workmanship

4.2.1.2 First article approval. First article approval shall be in accordance with the contract. (see 6.2.1g).

4.2.2 Acceptance inspection. Targets submitted for acceptance shall be capable of successfully passing the acceptance inspection specified in Table I as described in the following paragraphs to prove compliance with the requirements of 3.1.1, 3.2.2.1, 3.3.1, 3.3.2.3, 3.3.2.4, 3.3.3, 3.6 and 3.7.

4.2.2.1 Nose alignment. The nose cone shall be mounted on the fuselage and placed on a flat surface with the tow-point up. The distance from the surface to the most forward point on the nose cone shall be measured with a steel rule. The fuselage shall then be rolled 90 degrees plus or minus 2 degrees, and the measurement repeated. Both measurements shall be 6.0 $\pm$ 0.5 inch.

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TABLE I. Acceptance inspection check list.

Item	Examination for Target Serial No. ___	Inspector			
		Contractor		Government	
		Initials	Date	Initials	Date
1	Nose Alignment, (6.0 ± .5 in) Pitch Plane _____ Yaw Plane _____	_____	___	_____	___
2	Fin Twist (± .10 in) Fin 1 _____ Fin 2 _____ Fin 3 _____ Fin 4 _____	_____	___	_____	___
3	Fin Alignment (± .15 in) Pair 1 _____ Pair 2 _____	_____	___	_____	___
4	Fin tip-to-Fin tip (dihedral) (25.4 ± .8 in) Pair 1 _____ Pair 2 _____ Pair 3 _____ Pair 4 _____	_____	___	_____	___
5	CG (2.0 ± .5 in) _____	_____	___	_____	___
6	Strongback edge distance (.515 min) Right Side _____ Left Side _____	_____	___	_____	___
7	Towbolt Centerline to Fuselage Skin (.750 + .060/- .030) _____	_____	___	_____	___
8	Casting Integrity	_____	___	_____	___
9	Interchangeability and Maintainability Interchange two fins (10 minutes) _____ Exchange target noses (10 minutes) _____	_____	___	_____	___
10	Install Government property (Satisfactory, Unsatisfactory) TAS RACK _____ Radar Non-Reflective Nose Cone _____	_____	___	_____	___
11	Workmanship, Nameplates and Painting and Marking, International Orange with "TDU-34/A" (Satisfactory, Unsatisfactory) _____	_____	___	_____	___

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4.2.2.2 Fin twist. Fin twist shall be checked by laying the blade of each fin on a flat surface and measuring the out-of-flatness with feeler gages. The maximum gap between the fin blade and the flat surface shall be not greater than 0.10 inch.

4.2.2.3 Fin alignment. Pairs of opposing fins (that is 45° and 225°, 135° and 315°) shall be parallel to avoid introducing an aerodynamic torque during flight. Fin alignment shall be checked by installing all four fins and measuring parallelism on each opposing pair using feeler gages and a jig that straddles the fuselage and supports a pair of straight-edges that contact the root chords of the opposing fins. The maximum gap between the straight-edges and the fin surfaces shall be not greater than 0.15 inch.

4.2.2.4 Fin dihedral. The dihedral angle between each pair of adjacent fins shall be checked by measuring between the outboard fin tip to fin tip of each pair of adjacent fins with a steel rule. This distance shall be 25.4 $\pm$ 0.8 inch.

4.2.2.5 Center of gravity. The target shall be lifted with the CG Fixture, Drawing No. 573962, and the lifting strap shall be between the scribed lines on the fixture, which indicates an acceptable target center of gravity.

4.2.2.6 Strongback edge distance. The strongback tow-point edge distance shall be measured as shown in Figure 2, with calipers. This measurement plus one-half the hole diameter shall be not less than 0.515-inch for both webs.

4.2.2.7 Tow-bolt location. The distance from the centerline of the tow-bolt hole in the strongback to the upper surface of the body shall be 0.750 (+0.060, -0.030) inch. This distance shall be measured using the depth measurement feature of a set of calipers as shown in Figure 2. The caliper measurement shall be: 0.750 (+.060, -.030) minus the bolt diameter plus the hole diameter divided by 2 plus the rule width.

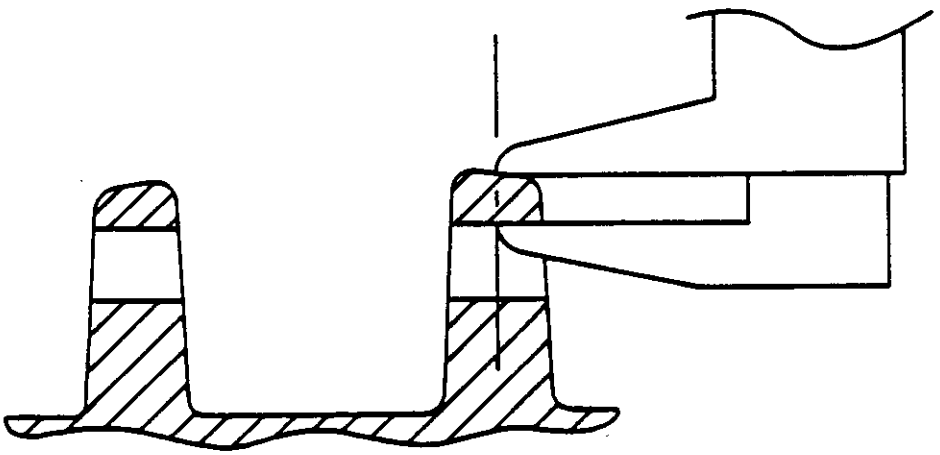
4.2.2.8 Casting integrity. Castings shall be inspected visually and the tow-point region of each strongback casting shall be dye penetrant inspected after drilling by a qualified inspection facility, using a fluorescent dye in accordance with MIL-C-6021.

4.2.2.9 Interchangeability and maintainability. To verify interchangeability (see 6.3.2) and maintainability, noses and fins shall be interchanged between two randomly selected targets, and the targets reassembled. The time to disassemble and reassemble a target nose or fin shall be not greater than 10 minutes.

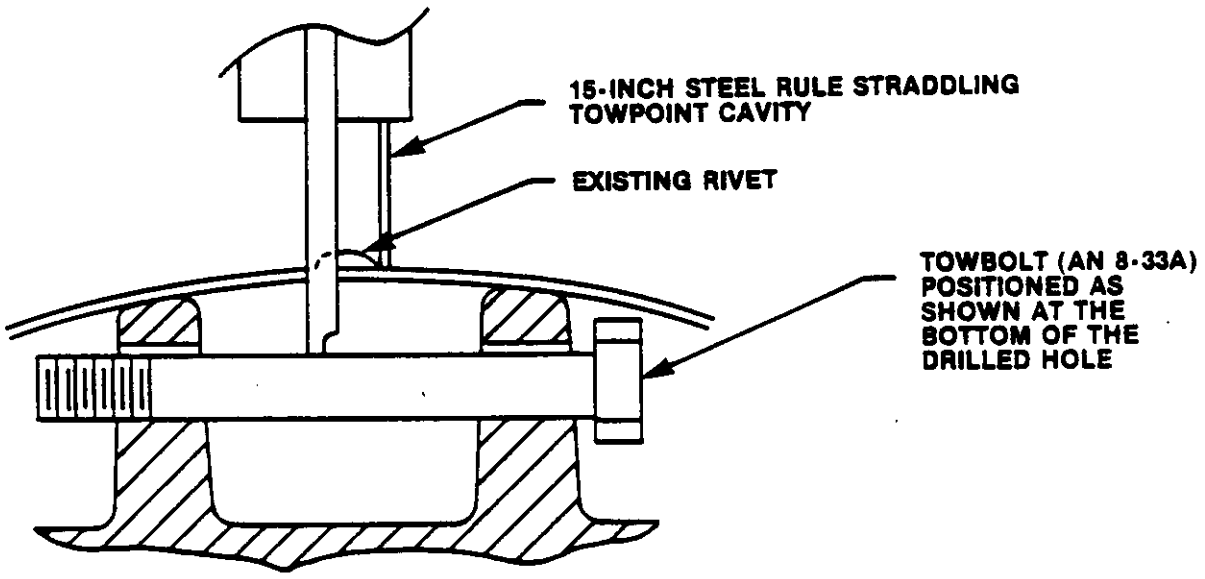
4.2.2.9.1 Interchangeability with government equipments. To verify interchangeability with government equipments, the Government-loaned (see 3.1.2) equipment shall be installed and removed from one target, unless otherwise specified in the contract to provide quality assurance.

4.2.2.10 Workmanship and other examinations. The targets shall be examined for workmanship, painting and marking, and nameplate.

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EDGE DISTANCE



TOW-BOLT LOCATION

FIGURE 2. Strongback inspection

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4.2.2.11 Methods of inspection. A sample of targets selected at random from each inspection lot shall be inspected in accordance with the procedures of 4.2.2.1 through 4.2.2.10. The results of these tests shall be recorded for each sample target on an acceptance inspection check off list similar to Table I which shall be retained at the contractor's facility. Unless otherwise specified in the contract, the size of the inspection lot shall be 100 targets (see 6.2.1c). The sample size and acceptance and rejection criteria shall be in accordance with MIL-STD-105 using an Acceptable Quality Level of 1.0 and General Inspection Level II (see 6.2.1h).

4.2.2.12 Acceptance approval. The acceptance approval will be by the Contract Administrative Officer.

4.2.3 Special inspections. Special inspections will be conducted by the Government when necessary (see 4.1) to assure that the required performance and physical characteristics (see 3.2) have been maintained by the contractor. Special tests will consist of any one or more or all of the following:

a. Flight demonstrations of Table II to assess the capability of the target to comply with the following:

   (1) Performance (see 3.2.1)
   (2) Flight (see 3.2.1.1)
   (3) Structural strength (see 3.2.2.2)
   (4) Operational environment (see 3.2.3.1)
   (5) Stowage environment (see 3.2.3.2)

b. Ground tests to assess the capability of the target to comply with the following:

   (1) Minimum radar cross section (see 3.2.1.2)
   (2) Survivability (see 3.2.2.3)
   (3) Tow system interface (see 3.3.2.6)

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging of the target for serviceability at destination and verification thereof shall be in accordance with MIL-STD-794, Level A (see 6.2.1d).

5.2 Packing. Packing for each target shall be Level A in accordance with MIL-STD-794, and shall be not less than a double wall weather resistant fiberboard box conforming to PPP-B-636, except neither a wooden nor a metal container shall be selected (see 6.2.1d)

5.3 Specific considerations.

5.3.1 Target disassembly. The four target fins, radar reflective nose cone, and target adapter assembly shall be disassembled and organized within the container to reduce shipping volume. Attaching parts shall be bagged or otherwise secured and identified for proper reassembly of the target.

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TABLE II. Flight tests

Test No. 1	Test No. 2	Test No. 3	Test No. 4
1. Launch and deploy 1,000 ft of towline at 10,000 ft altitude & 300 KCAS airspeed (1000 fpm max reel-out speed)	1. Same as one in Test No. 1, then increase speed to 350 KCAS.	1. Launch and deploy 1,000 ft of towline at 30,000 ft altitude & 260 KCAS airspeed (1000 fpm max reel-out speed)	1. Launch and deploy 30,000 ft of towline at 260 KCAS and 10,000 ft altitude (3000 fpm max reel out speed)
2. Assess stability at following airspeeds: 260, 300, 350, 400, 450 & 570 KCAS.	2. Assess maneuverability using 180° 2g PORT & STBD turns at 350 & 400 KCAS.	2. Recover at same altitude and airspeed.	2. Tow straight & level at the following airspeeds: 260, 300, 325, & 350 KCAS.
3. Assess stability with 20,000 feet of towline at 260, 300 & 350 KCAS airspeed.	3. Recover target at 300 KCAS airspeed & 10,000 feet altitude.	3. Repeat 1. & 2. with 1,500 ft of towline at the following KCAS/altitudes: 300/30,000 260/5,000 300/5,000	3. Repeat 4. of Test No. 1.
4. Recover target at 260 KCAS airspeed & 10,000 feet altitude.	4. Repeat 1. with 500 feet of towline, then repeat 3. above.		4. Repeat 5. of Test No. 1.
5. Repeat 1. with 500 ft of towline at 260 KCAS, then repeat 4.	5. Increase airspeed to 400 KCAS and perform 3g PORT and STBD turns.		

NOTES: 1. Evaluate stowed target during catapult take-off and arrested landings of aircraft.
2. Test values are to be interpreted as "not less than" minimums.

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5.3.2 Container stacking. The container structural design shall provide rigidity for storage in stacks of six containers.

5.3.3 Verified packaging and packing. Drawing 573011 discloses a container which has been verified for compliance with the preservation-packaging and packing requirements herein. Drawing changes or superior substitutes shall be approved by the Contract Administrative Officer (see 6.2.1e).

5.4 Marking. Marking for shipment shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The target is intended to be installed on an aircraft reeling machine, streamed and flown as a tow target for surface-to-air or air-to-air gunnery or missile training, recovered by the aircraft reeling machine, and landed with the aircraft aboard a carrier or at a land base.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Current top assembly drawing for the target (see 3.3.1).

c. Inspection lot size, if other than specified in 4.2.2.11.

d. Levels of preservation and packaging and packing (see 5.1 and 5.2).

e. Configuration control procedures in accordance with DOD-STD-480 for the target drawings. Targets previously fabricated in accordance with drawing package 573850 have been demonstrated to comply with the requirements herein. Accordingly, proposed changes to the drawings may require contractor analyses and Government verification to confirm that the requirements herein will not be degraded (see 3.2.1).

f. Authorization to use the Government-loaned property listed in 3.1.2.

g. Specific instructions in acquisition documents regarding arrangements for inspection and approval of the documents first article. When a first article inspection is required, the item will be tested and should be a first article sample. The first article should consist of 10 units (see 4.2.1).

h. The manner in which rejected targets may be repaired or corrected and resubmitted for approval. Unacceptable lots should be resubmitted for acceptance inspection after contractor inspection of 100 percent of the targets in the lot, and after all defective targets have been removed or corrected. The method of reinspection should be determined by the Contract Administrative Officer in accordance with MIL-STD-105 (see 4.2.2.11).

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6.3 Definitions.

6.3.1 Complete provisions. Complete provisions shall be defined to mean that all supports, brackets, tubes, and fittings, etc., have been installed and adequate weight and space allowed in order that the equipment can be installed with or without ballast alteration to the target, and that no additional parts are required for installation, other than the item itself. Standard stock items such as nuts, bolts, cotter pins, etc., need not be furnished.

6.3.2 Interchangeable. When two or more items possess such functional and physical characteristics as to be equivalent in performance and durability and capable of being exchanged one for the other without alteration of the items themselves, or of adjoining items except for adjustment, and without selection for fit or performance, the items are interchangeable. (See DoD-STD-100B).

Preparing activity:
Navy-AS
(Project No. 1550-N003)

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APPENDIX A

RADAR CROSS SECTION REQUIREMENTS FOR THE TDU-34/A TOW TARGET

10. SCOPE

10.1 Scope. This appendix presents the monostatic radar cross section (RCS) signature to be exhibited by the TDU-34/A aerial tow target. This Appendix is a mandatory part of the specification. The information contained herein is intended for compliance. Any change to the target design which may result in a change in the RCS signature constitutes a change in function and requires engineering change proposal documentation and technical evaluation for Government approval (see 6.2.1e).

20. APPICABLE DOCUMENTS

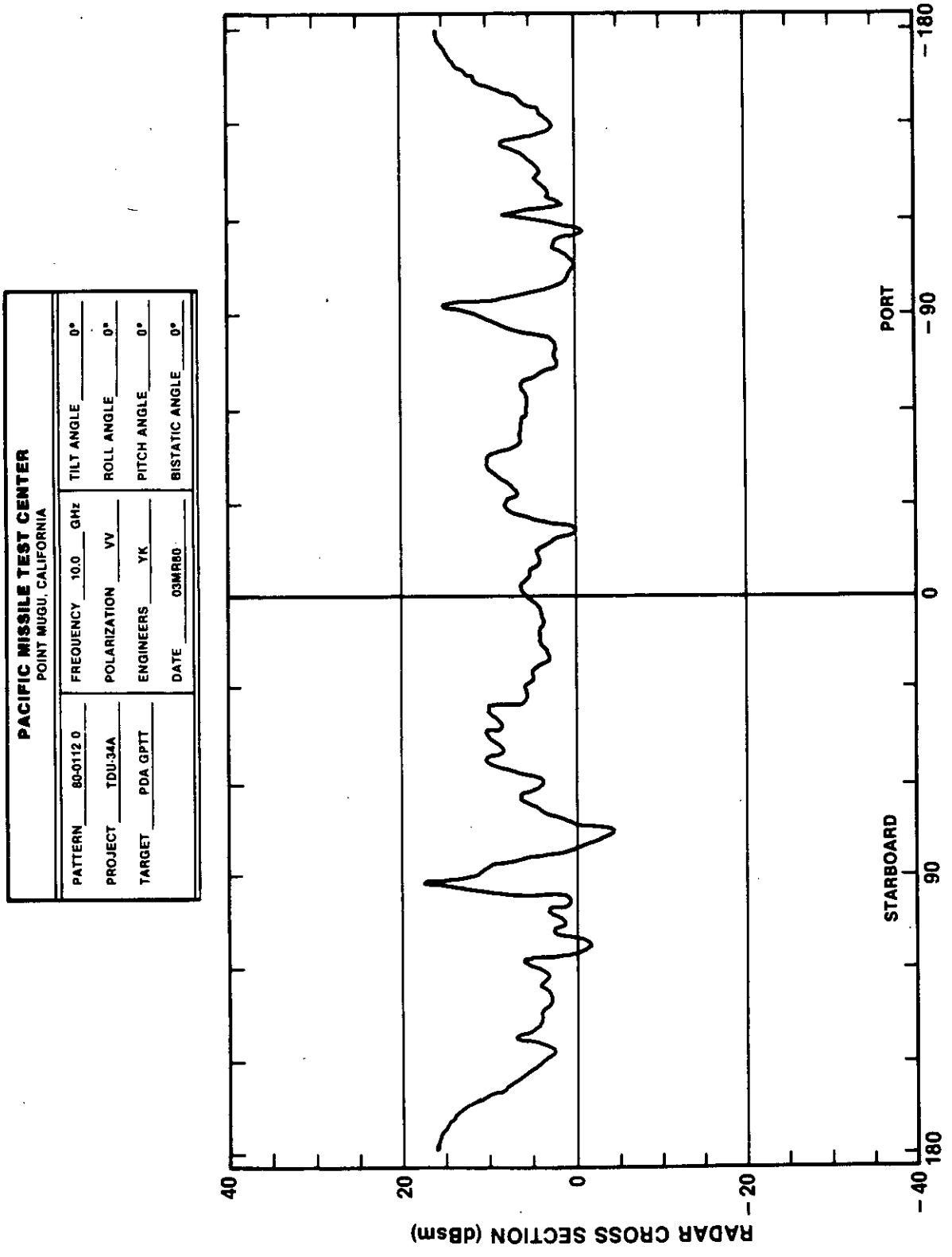
(This section not applicable to this appendix.)

30. REQUIREMENTS

30.1 Radar Cross Section. When irradiated by any monstatic, linearly polarized radar operating at 10.0 Gigahertz (GHz), the target shall exhibit median RCS signatures similar to those of figures A-1 through A-10. From a head-on view within a 30 degree conical angle, the target shall have a minimum of one square meter reflective area. For all other aspect angles, maxima (peaks) in the median RCS signature greater than or equal to 8 dBsm and minima (nulls) less than or equal to 0 dBsm shall occur within $\pm$5 degrees aspect angle of their associated maxima and minima in figures A-1 through A-10. The target median RCS shall be within $\pm$1 dBsm of the median RCS of figures A-1 through A-10 in a $\pm$5 degrees aspect angle band about each maxima and minima so defined. All other points on the target median RCS signature shall be within $\pm$1 dBsm of the associated points on figures A-1 through A-10.

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APPENDIX A

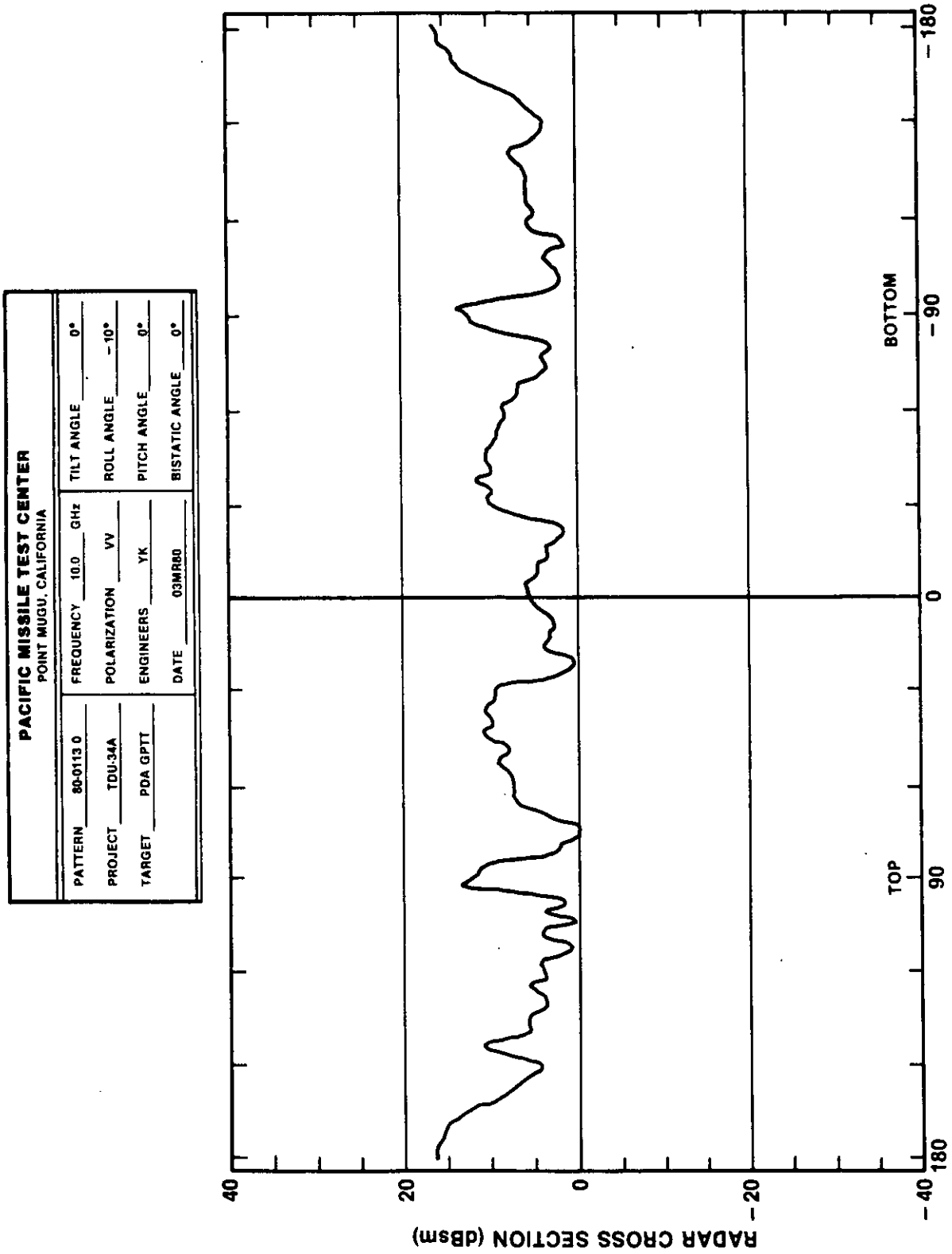


RCS FOR 0° ROLL ANGLE

FIGURE A-1

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APPENDIX A

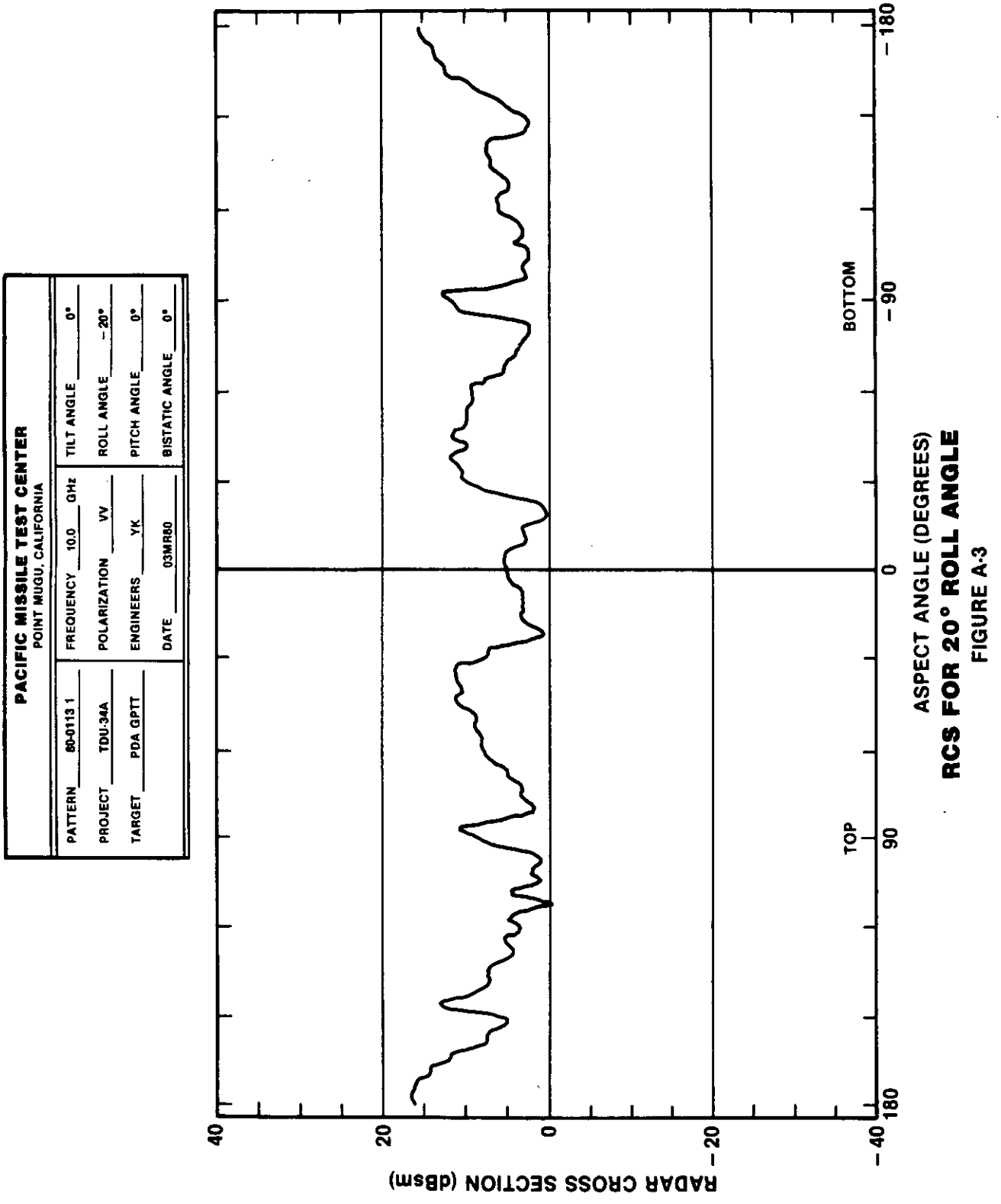


RCS FOR 10° ROLL ANGLE

FIGURE A-2

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APPENDIX A

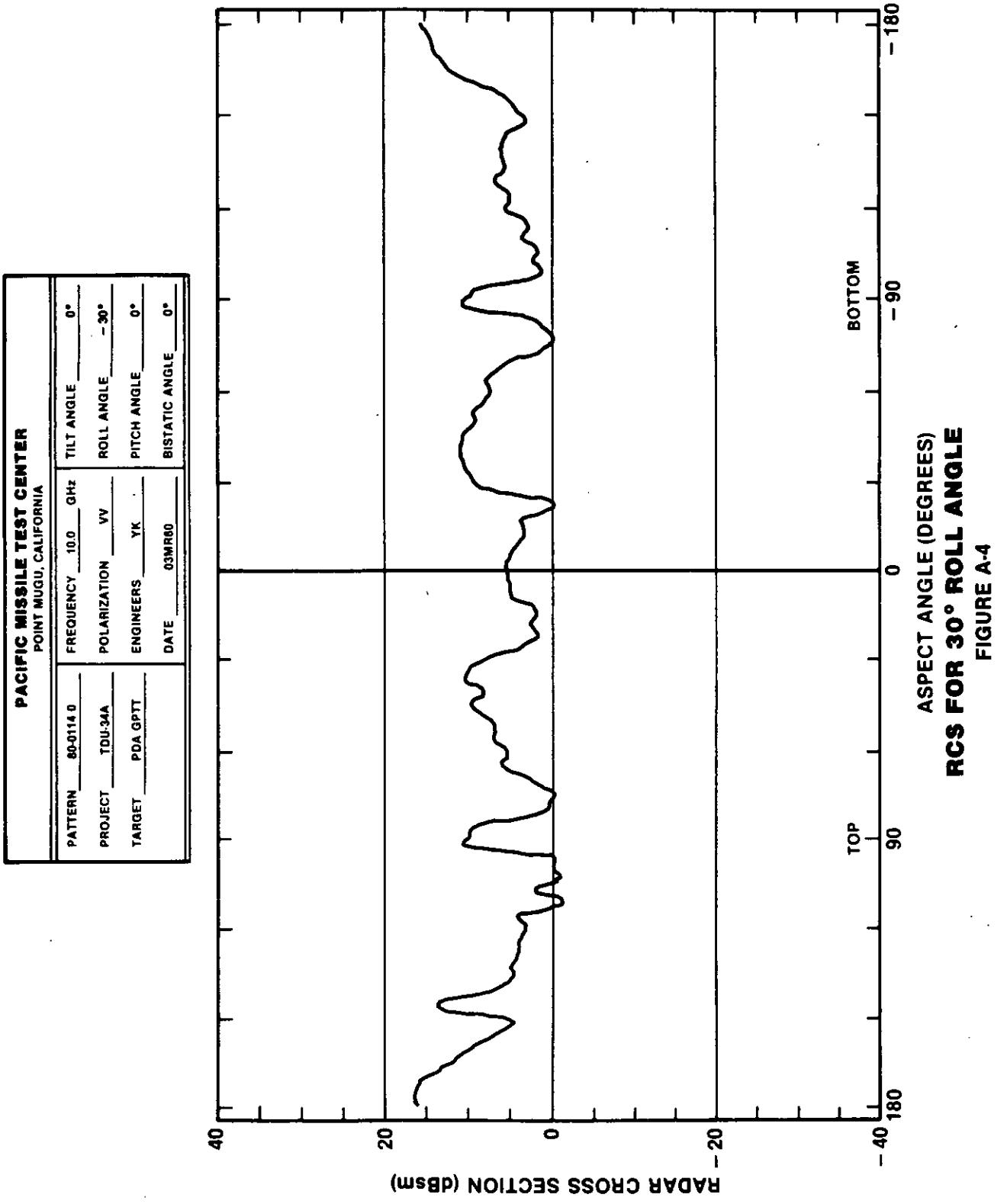


RCS FOR 20° ROLL ANGLE

FIGURE A-3

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APPENDIX A

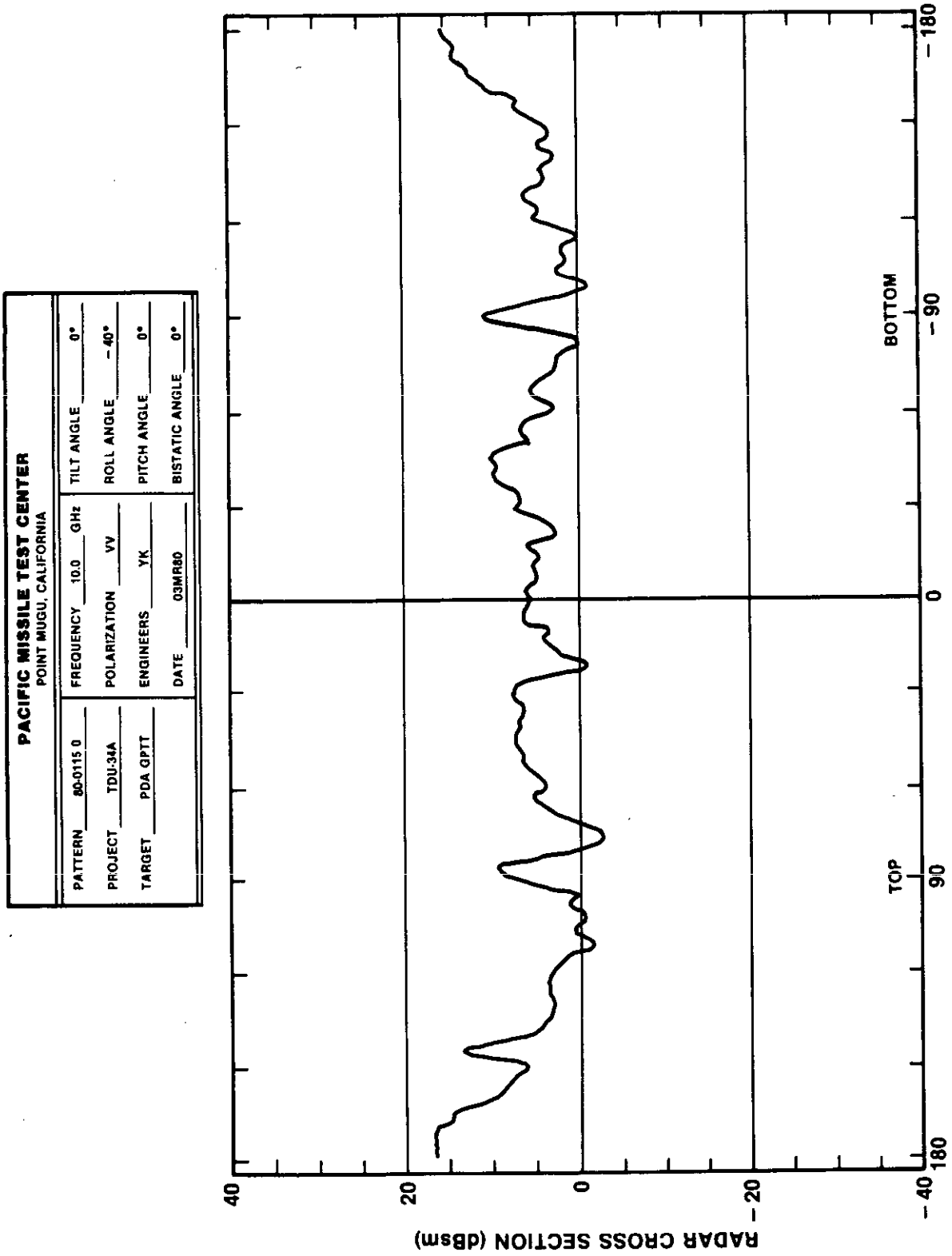


RCS FOR 30° ROLL ANGLE

FIGURE A-4

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APPENDIX A

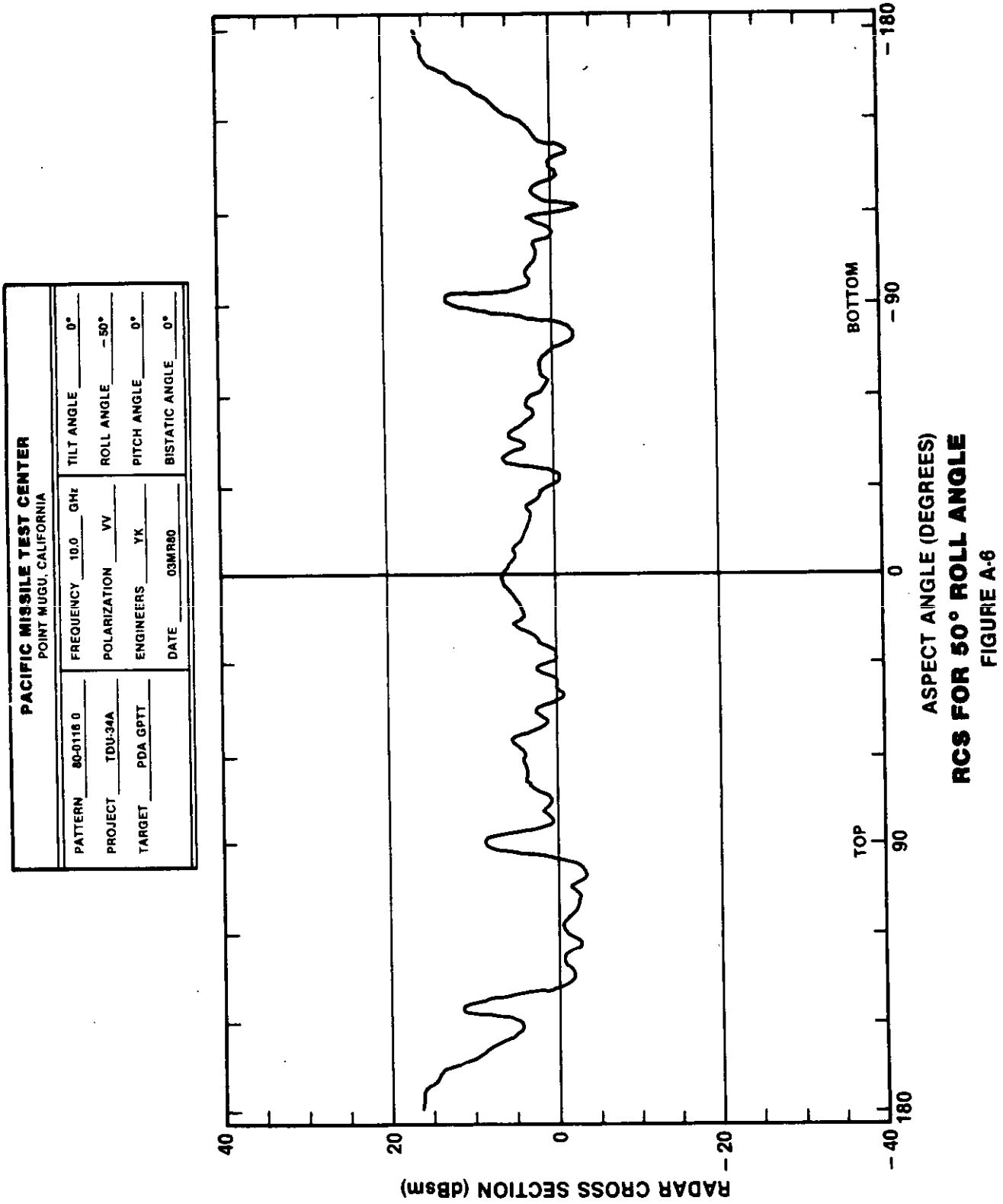


RCS FOR 40° ROLL ANGLE

FIGURE A-5

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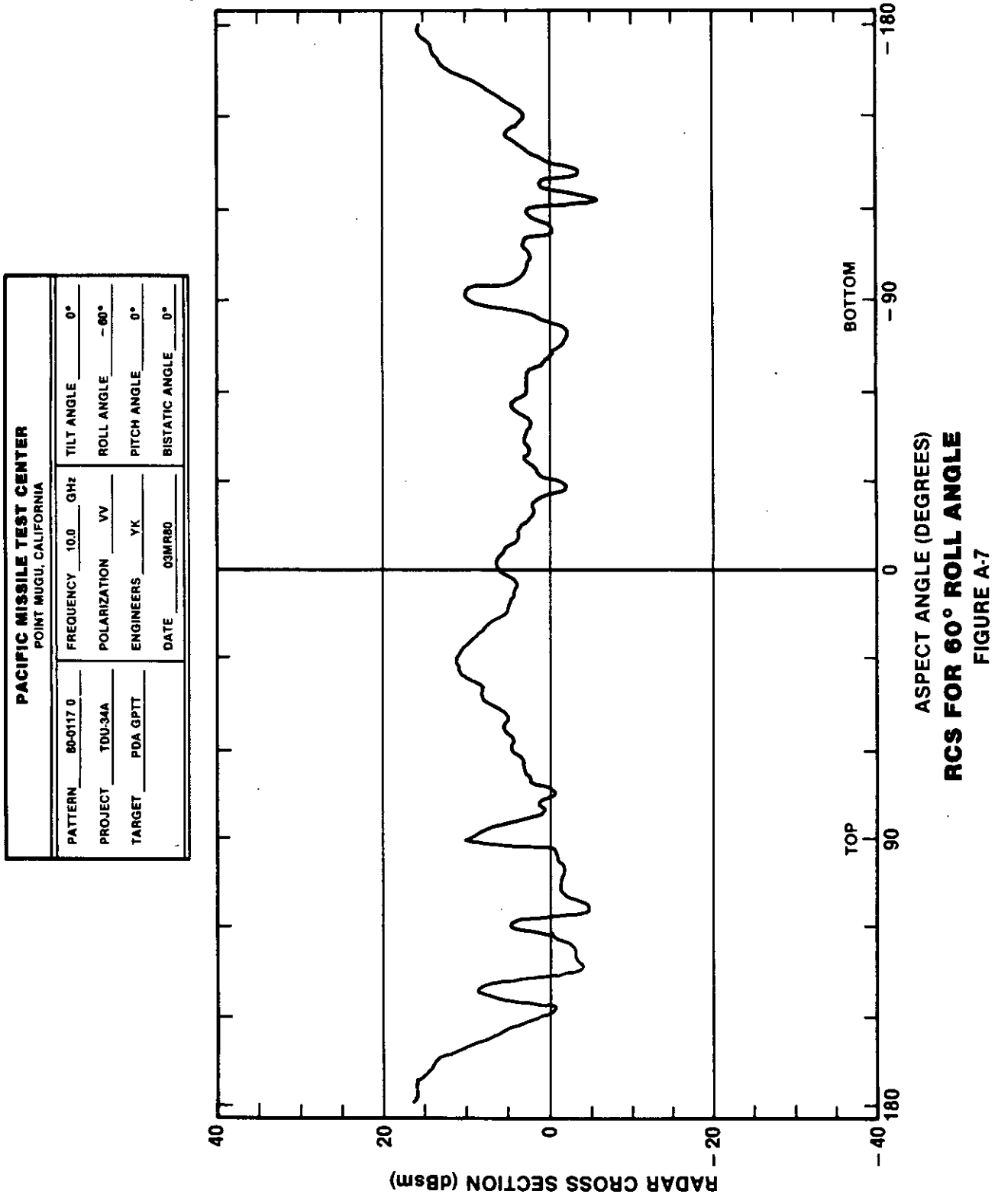


RCS FOR 50° ROLL ANGLE

FIGURE A-6

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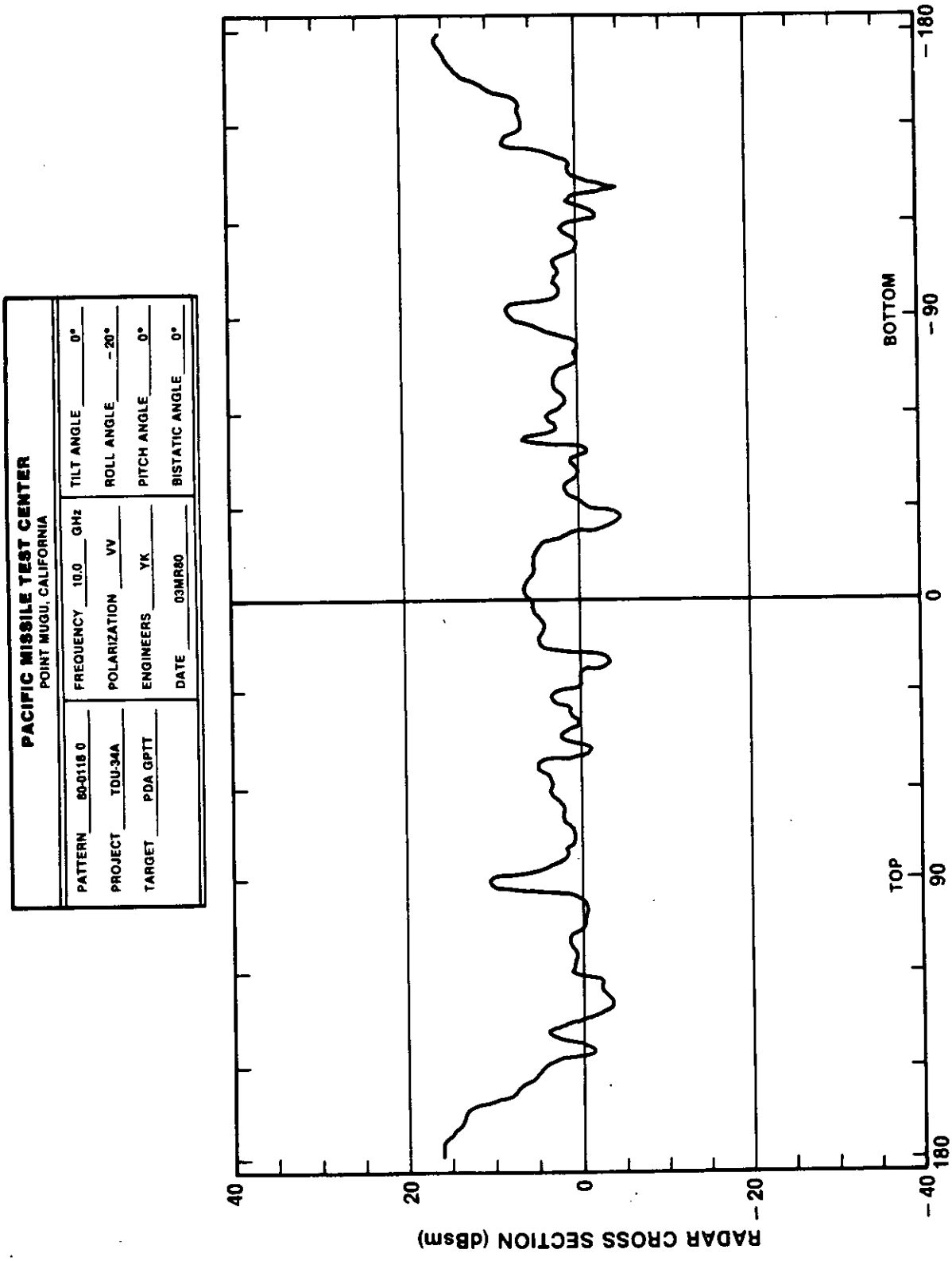


RCS FOR 60° ROLL ANGLE

FIGURE A-7

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APPENDIX A

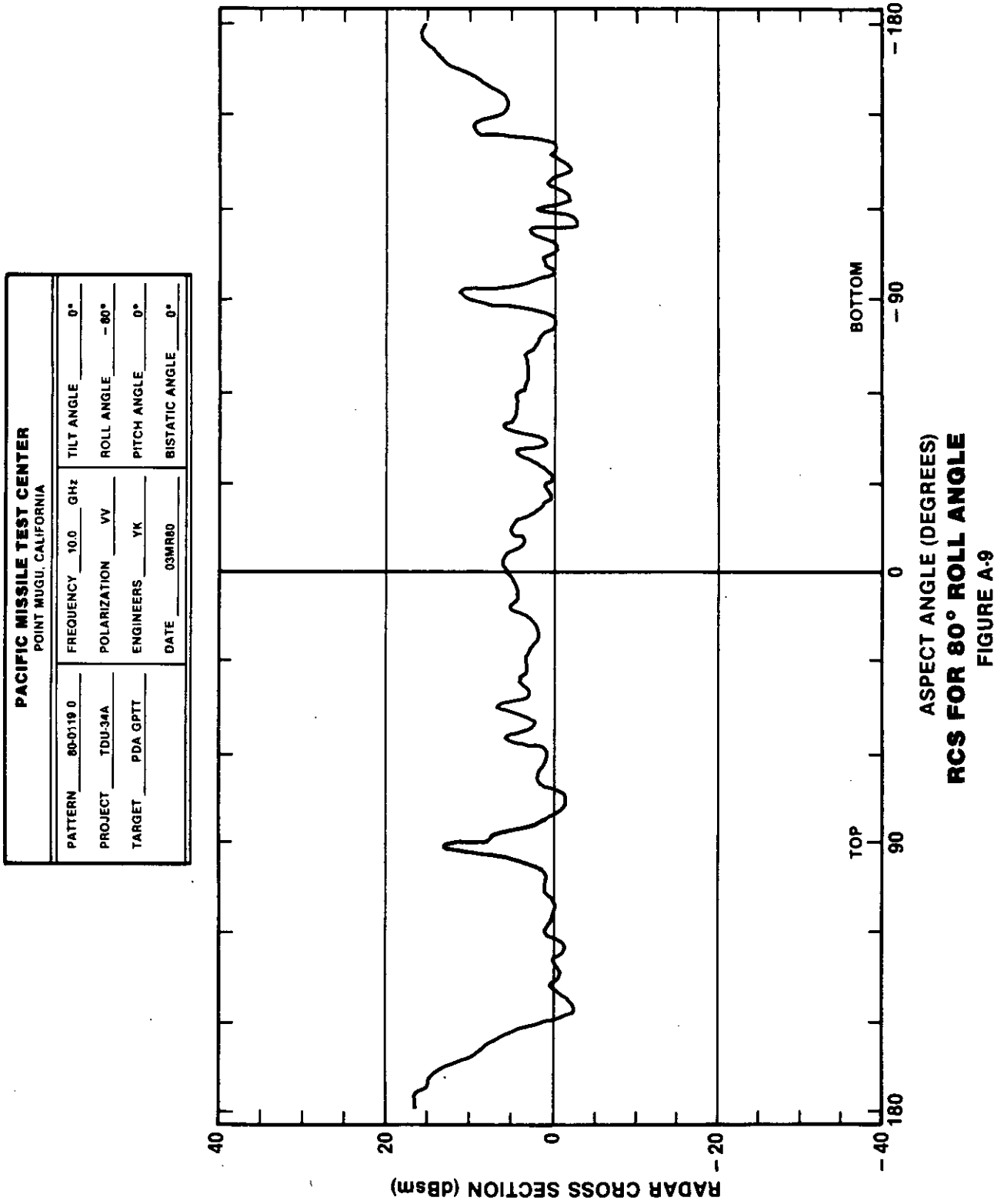


RCS FOR 70° ROLL ANGLE

FIGURE A-8

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APPENDIX A

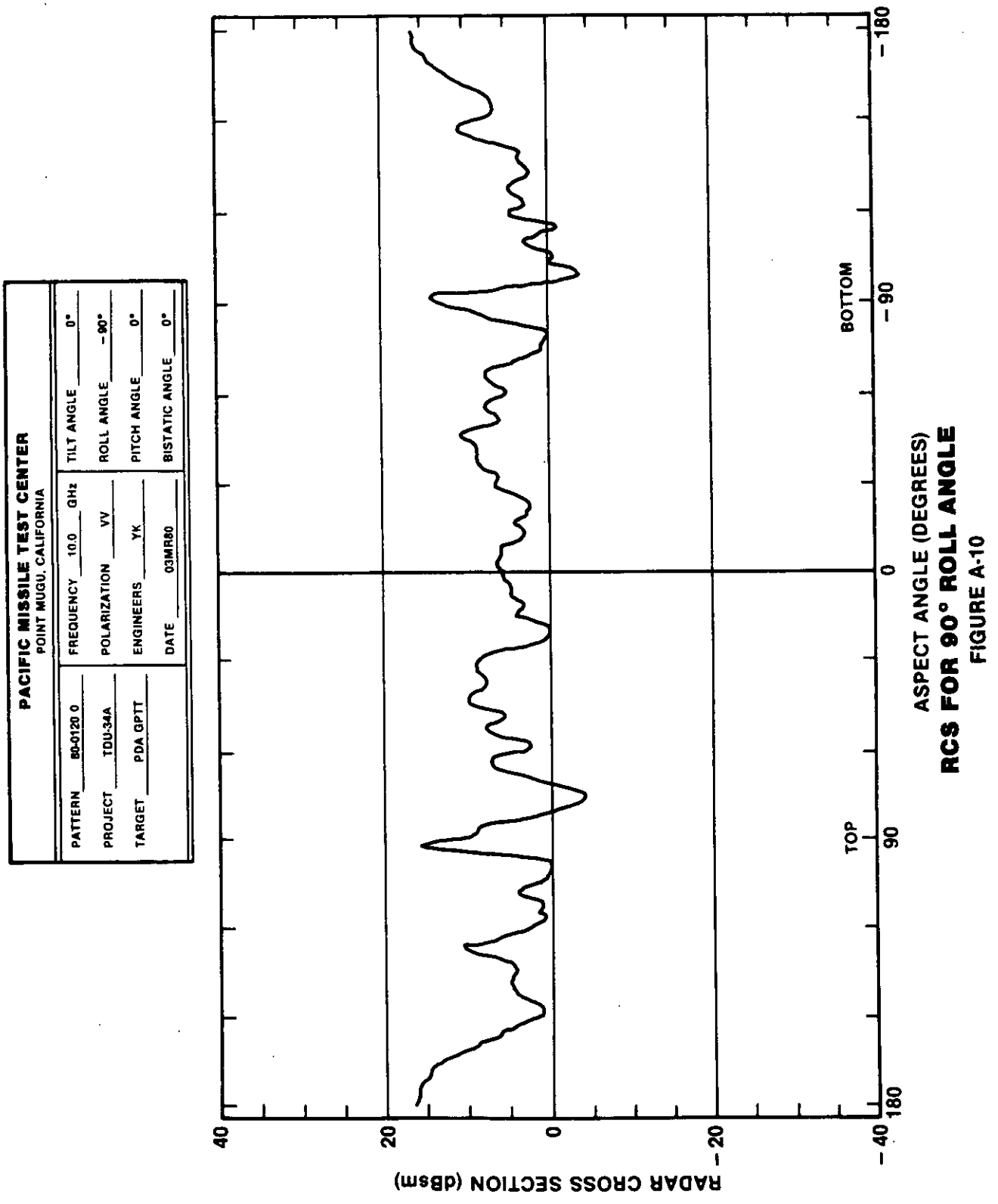


RCS FOR 80° ROLL ANGLE

FIGURE A-9

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APPENDIX A



RCS FOR 90° ROLL ANGLE

FIGURE A-10

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APPENDIX B

METHOD FOR COMPUTING LOADS INDUCED BY AIRCRAFT CATAPULT, ARRESTMENT AND FLIGHT ON A STOWED TOW TARGET

10. SCOPE

10.1 Scope. This appendix outlines a computational program which shall be used to transition the design load criteria of MIL-A-8591 to a center-of-gravity tow target which is engaged in a tow target reeling machine-launcher (the airborne store) by the target adapter assembly, and is retrained within the tow target reeling machine-launcher by four launcher pads that interface with the target fuselage. This Appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS. This section not applicable to this appendix.

30. METHOD

30.1 Tow target program. The program outlined herein shall be used to compute the loads induced on a tow target during aircraft catapult, arrestment and flight; and for designing the target fuselage and structure to withstand the induced aircraft loads on the tow target when stowed in the tow target reeling machine-launcher.

```
      PROGRAM STORE (INPUT, OUTPUT, TAPE 5 = INPUT, TAPE 6 = OUTPUT)
C
C     THE ABOVE PROGRAM CARD IS FOR CDC COMPUTER SYSTEM USE.  WHEN USING
C     OTHER THAN CDC SYSTEMS, THE PROGRAM CARD MUST BE DELETED.
C
      DIMENSION TITLE (17)
C
C     SIGN CONVENTIONS LOADS      PX  + AFT
C                                 PY  + LEFT LOOKING FORWARD
C                                 PZ  + UP
C                                 AMX + COUNTERCLOCKWISE LOOKING FORWARD
C                                 AMY + NOSE UP
C                                 AMZ + NOSE LEFT LOOKING FORWARD
C                    LUG REACTIONS + FORWARD, LEFT LOOKING FORWARD, UP
C                    SWAYBRACE REACTIONS + INTO STORE
C
C     INPUT DEFINITIONS
C         DEL-RACK MOUNTING ANGLE (FROM VERTICAL),DEG
C         ALF-DISTANCE FROM CG TO FWD LUG,IN
C         ALA-DISTANCE FROM CG TO AFT LUG,IN
C         ALBF-DISTANCE FROM CG TO FWD SWAY BRACE,IN
C         ALBA-DISTANCE FROM CG TO AFT SWAY BRACE,IN
C         BETAF-FWD SWAY BRACE ANGLE,DEG
C         BETAA-AFT SWAY BRACE ANGLE,DEG
C         SB-LOAD FRACTION OF YM REACTED BY SWAY BRACE (.4 USUALLY ASSUMED)
C         W-WEIGHT OF STORE,LB
```


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```
C         AI-MASS MOMENT OF INERTIA,LB-SQ IN
C         R-RADIUS OF STORE,IN
C         C-LUG HEIGHT (TO REACT FORE AND AFT LOADS),IN
C         H-LUG HEIGHT (TO REACT SIDE LOADS),IN
C         E1-VERT DIST BETWEEN GEO CENTER & SWAY BRACE ACTION LINE(POS UP),IN
C         E2-VERT DIST FROM GEO CENTER TO CG (POS DWN),IN
C         E3-LAT DIST FROM GEO CENTER TO CG (POS TO RIGHT LOOKING FWD),IN
C         ANX-AXIAL LOAD FACTOR
C         ANY-LAT LOAD FACTOR
C         ANZ-NORMAL LOAD FACTOR
C         THEDD-PITCHING ACCELERATION
C         PSIDD-YAWING ACCELERATION
C         PXA-AXIAL AIR LOAD,LB
C         PYA-LAT AIR LOAD,LB
C         PZA-NORMAL AIR LOAD,LB
C         AMXA-ROLLING MOMENT,IN LB
C         AMYA-PITCHING MOMENT,IN LB
C         AMZA-YAWING MOMENT,IN LB
C         SINGLE LUG PROBLEMS
C         ALF IS POSITIVE IF THE LUG IS AFT OF THE CG
C         ALA IS SET EQUAL TO ZERO
C    READ TITLE CARD
C
          READ(5,1000)NLUGS, IPRINT,TITLE
C
C    READ LAUNCHER GEOMETRY
C
          READ(5,1001)DEL,ALF,ALA,ALBF,ALBA,BETAF,BETAA,SB
C
C    READ STORE CHARACTERISTICS
C
          READ(5,1001)W,AI,R,C,H,E1,E2,E3
C
C    CALCULATE INTERMEDIATE VARIABLES
C

          SIND=SIN(DEL/57.3)
          COSD=COS(DEL/57.3)
          SINBF=SIN(BETAF/57.3)
          COSBF=COS(BETAF/57.3)
          TANBF=SINBF/COSBF
          SINBA=SIN(BETAA/57.3)
          COSBA=COS(BETAA/57.3)
          TANBA=SINBA/COSBA
C
C    PRINT GEOMETRY AND CHARACTERISTICS
C

          1 IF(NLUGS.EQ.1)GO TO 2
            WRITE(6,2000)TITLE
```


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```
      WRITE(6,2001)W,AI,ALF,R,E1,ALA,C,E2,ALBF,H,E3,ALBA,BETAF,SB,BETAA,
      X          DEL
      WRITE(6,2002)
      N=0
      GO TO 3
    2 AL=ALF
      WRITE(6,2000)TITLE
      WRITE(6,2003)W,AI,AL,R,E1,C,E2,ALBF,H,E3,ALBA,BETAF,BETAA,DEL
      WRITE(6,2002)
      N=0
C
C   READ FLIGHT CONDITIONS
C
    3 READ(5,1002)NCOND,ANX,ANYO,ANZ,THEDDO,PSIDDO
C
C   TEST FOR END OF RUN
C
      IF(ANX)4,69,4
C
    4 READ(5,1001)PXA,PYA,PZA,AMXA,AMYA,AMZA
      N=N+1
C
C   TRANSFORM APPLIED LOADS INTO MISSILE COORDINATE SYSTEM
C
      PX=PXA+ANX*W
      PY=(PYA+ANY*W)*COSD-(PZA=ANZ*W)*SIND
      PZ=(PYA+ANY*W)*SIND+(PZA=ANZ*W)*COSD
      AMX=AMXA-(ANY*W*COSD-ANZ*W*SIND)*E2+(ANY*W*SIND+ANZ*W*COSD)*E3
      AMY=(AMYA+AI*THEDD/386.)*COSD+(AMZA+AI*PSIDD/386.)*SIND-ANX*W*E2
      AMZ=-(AMYA+AI*THEDD/386.)*SIND+(AMZA+AI*PSIDD/386.)*COSD-ANX*W*E3
      IF(NLUGS-1)5,38,5
C
C   TANDEM LUG CONFIGURATION
C
C   CALCULATE VERTICAL COMPONENTS OF SWAYBRACE REACTIONS TO PY, AMX,
C   AND AMZ
C
    5 VBF=((PY*(R+H)-AMX)*ALBA+SB*AMZ*(R+H-E1))/(R+H-E1)/(ALBF+ALBA)/
      X     TANBF
      VBA=((PY*(R+H)-AMX)*ALBF-SB*AMZ*(R+H-E1))/(R+H-E1)/(ALBF+ALBA)/
      X     TANBA
      IF(VBF)6,7,7
    6 IBF=-1
      VBF=-VBF
      GO TO 8
    7 IFB=1
    8 IF(VBA)9,10,10
    9 IAB=-1
      VBA=-VBA
      GO TO 11
   10 IAB=1
```


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APPENDIX B

```
C
C    CALCULATE TRIAL LUG REACTIONS
C
        11 RZFLT=(PX*(R+C)-PZ*ALA-AMY+VBF*(ALBF+ALA)+VBA*(ALA-ALBA))/
         X       (ALF+ALA)
           RZALT=(-PX*(R+C)-PZ*ALF+AMY+VBF*(ALF-ALBF)+VBA*(ALF+ALBA))/
         X       (ALF+ALA)
C
C    DETERMINE APPLICABLE CASE
C
           IF(RZFLT)12,13,13
        12 IF(RZALT)32,26,26
        13 IF(RZALT)20,14,14
C
C    CASE I FORWARD AND AFT LUGS
C
        14 RZFL=RZFLT
           RZAL=RZALT
           RYFL=((PY*E1-AMX)*ALA-(1.-SB)*AMZ*(R+H-E1))/(ALF+ALA)/(R+H-E1)
           RYAL=((PY*E1-AMX)*ALF+(1.-SB)*AMZ*(R+H-E1))/(ALF+ALA)/(R+H-E1)
           RXL=PX
           IF(IFB)15,16,16
        15 RFLB=0.
           RFRB=VBF/COSBF
           GO TO 17
        16 RFLB=VBF/COS(BETAF/57.3)
           RFRB=0.
        17 IF(IAB)18,19,19
        18 RALB=0.
           RARB=VBA/COSBA
           GO TO 65
        19 RALB=VBA/COSBA
           RARB=0.
           GO TO 65
C
C    CASE IIA FORWARD LUG AND AFT SWAYBRACES
C
        20 RZFL=(PX*(R+C)-PZ*ALBA-AMY+VBF*(ALBF+ALBA))/(ALF+ALBA)
           RZAL=0.
           RYFL=((PY*E1-AMX)*ALA-(1.-SB)*AMZ*(R+H-E1))/(ALF+ALA)/(R+H-E1)
           RYAL=((PY*E1-AMX)*ALF+(1.-SB)*AMZ*(R+H-E1))/(ALF+ALA)/(R+H-E1)
           RXL=PX
           IF(IFB)21,22,22
        21 RFLB=0.
           RFRB=VBF/COSBF
           GO TO 23
        22 RFLB=VBF/COSBF
           RFRB=0.
        23 IF(IAB)24,25,25
        24 RALB=(PX*(R+C)+PZ*ALF-AMY+VBF*(ALBF-ALF)-VBA*(ALF+ALBA))/2./
           X    (ALF+ALBA)/COSBA
           RARB=(PX*(R+C)+PZ*ALF-AMY+VBF*(ALBF-ALF)+VBA*(ALF+ALBA))/2./
```


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```
           X    (ALF+ALBA)/COSBA
           GO TO 65
        25 RALB=(PX*(R+C)+PZ*ALF-AMY+VBF*(ALBF-ALF)+VBA*(ALF+ALBA))/2./
           X    (ALF+ALBA)/COSBA
           RARB=(PX*(R+C)+PZ*ALF-AMY+VBF*(ALBF-ALF)-VBA*(ALF+ALBA))/2./
           X    (ALF+ALBA)/COSBA
           GO TO 65
C
C   CASE IIB FORWARD SWAYBRACES AND AFT LUG
C
        26 RZFL=0.
           RZAL=(-PX*(R+C)-PZ*ALBF+AMY+VBA*(ALBF+ALBA))/(ALBF+ALA)
           RYFL=((PY*E1-AMX)*ALA-(1.-SB)*AMZ*(R+H-E1))/(ALF+ALA)/(R+H-E1)
           RYAL=((PY*E1-AMX)*ALF+(1.-SB)*AMZ*(R+H-E1))/(ALF+ALA)/(R+H-E1)
           RXL=PX
           IF(IFB)27,28,28
        27 RFLB=(-PX*(R+C)+PZ*ALA+AMY-VBF*(ALBF+ALA)+VBA*(ALBA-ALA))/2./
           X    (ALBF+ALA)/COSBF
           RFRB=(-PX*(R+C)+PZ*ALA+AMY+VBF*(ALBF+ALA)+VBA*(ALBA-ALA))/2./
           X    (ALBF+ALA)/COSBF
           GO TO 29
        28 RFLB=(-PX*(R+C)+PZ*ALA+AMY+VBF*(ALBF+ALA)+VBA*(ALBA-ALA))/2./
           X    (ALBF+ALA)/COSBF
           RFRB=(-PX*(R+C)+PZ*ALA+AMY-VBF*(ALBF+ALA)+VBA*(ALBA-ALA))/2./
           X    (ALBF+ALA)/COSBF
        29 IF(IAB)30,31,31
        30 RALB=0.
           RARB=VBA/COSBA
           GO TO 65
        31 RALB=VBA/COSBA
           RARB=0.
           GO TO 65
C
C   CASE III FORWARD AND AFT SWAYBRACES
C
        32 RZFL=0.
           RZAL=0.
           RYFL=((PY*E1-AMX)*ALA-(1.-SB)*AMZ*(R+H-E1))/(ALF+ALA)/(R+H-E1)
           RYAL=((PY*E1-AMX)*ALF+(1.-SB)*AMZ*(R+H-E1))/(ALF+ALA)/(R+H-E1)
           RXL=PX
           IF(IFB)33,34,34
        33 RFLB=(-PX*(R+C)+PZ*ALBA+AMY-VBF*(ALBF+ALBA))/2./(ALBF+ALBA)/COSBF
           RFRB=(-PX*(R+C)+PZ*ALBA+AMY+VBF*(ALBF+ALBA))/2./(ALBF+ALBA)/COSBF
           GO TO 35
        34 RFLB=(-PX*(R+C)+PZ*ALBA+AMY+VBF*(ALBF+ALBA))/2./(ALBF+ALBA)/COSBF
           RFRB=(-PX*(R+C)+PZ*ALBA+AMY-VFB*(ALBF+ALBA))/2./(ALBF+ALBA)/COSBF
        35 IF(IAB)36,37,37
        36 RALB=(PX*(R+C)+PZ*ALBF-AMY-VBA*(ALBF+ALBA))/2./(ALBF+ALBA)/COSBA
           RARB=(PX*(R+C)+PZ*ALBF-AMY+VBA*(ALBF+ALBA))/2./(ALBF+ALBA)/COSBA
           GO TO 65
        37 RALB=(PX*(R+C)+PZ*ALBF-AMY+VBA*(ALBF+ALBA))/2./(ALBF+ALBA)/COSBA
      RARB=(PX*(R+C)+PZ*ALBF-AMY-VBA*(ALBF+ALBA))/2./(ALBF+ALBA)/COSBA
```


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```
      GO TO 65
C
C   SINGLE LUG CONFIGURATION
C
C   CALCULATE VERTICAL COMPONENTS OF SWAYBRACE REACTIONS TO PY,AMX,
C   AND AMZ
C       38 VBF=(PY*(R+H)*ALBA-AMX*(ALBA-AL)+AMZ*(R+H-E1))/(R+H-E1)/
          X   (ALBF+ALBA)/TANBF
          VBA=(PY*(R+H)*ALBF-AMX*(ALBF+AL)-AMZ*(R+H-E1))/(R+H-E1)/
          X   (ALBF+ALBA)/TANBA
          IF(VBF)39,40,40
       39 IFB=-1
          VBF=-VBF
          GO TO 41
       40 IFB=1
       41 IF(VBA)42,43,43
       42 IAB=-1
          VBA=-VBA
          GO TO 44
       43 IAB=1
C
C   CALCULATE TRIAL LUG REACTIONS
C
       44 RZLTA=(PX*(R+C)-PZ*ALBA-AMY+VBF*(ALBF+ALBA))/(ALBA-AL)
          RZLTB=(-PX*(R+C)-PZ*ALBF+AMY+VBA*(ALBF+ALBA))/(ALBF+AL)
C
C   DETERMINE APPLICABLE CASE
C
          IF(RZLTA)45,45,46
       45 IF(RZLTB)59,59,46
       46 IF(RZLTA-RZLTB)53,53,47
C
C   CASE IIA LUG AND AFT SWAYBRACES
C
       47 RZL=RZLTA
          RYL=(PY*E1-AMX)/(R+H-E1)
          RXL=PX
          IF(IFB)48,49,49
       48 RFLB=0.
          RFRB=VBF/COSBF
          GO TO 50
       49 RFLB=VBF/COSBF
          RFRB=0.
       50 IF(IAB)51,52,52
       51 RALB=(PX*(R+C)-PZ*AL-AMY+VBF*(ALBF+AL)-VBA*(ALBA-AL))/2./
          X    (ALBA-AL)/COSBA
          RARB=(PX*(R+C)-PZ*AL-AMY+VBF*(ALBF+AL)+VBA*(ALBA-AL))/2./
          X    (ALBA-AL)/COSBA
          GO TO 65
       52 RALB=(PX*(R+C)-PZ*AL-AMY+VBF*(ALBF+AL)+VBA*(ALBA-AL))/2./
          X    (ALBA-AL)/COSBA
          RARB=(PX*(R+C)-PZ*AL-AMY+VBF*(ALBF+AL)-VBA*(ALBA-AL))/2./
```


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```
           X    (ALBA-AL)/COSBA
           GO TO 65
C
C    CASE IIB LUG AND FORWARD SWAYBRACES
C
        53 RZL=RZLTB
           RYL=(PY*E1-AMX)/(R+H-E1)
           RXL=PX
           IF(IFB)54,55,55
        54 RFLB=(-PX*(R+C)+PZ*AL+AMY-VBF*(ALBF+AL)+VBA*(ALBA-AL))/2./
           X    (ALBF+AL)/COSBF
           RFRB=(-PX*(R+C)+PZ*AL+AMY+VBF*(ALBF+AL)+VBA*(ALBA-AL))/2./
           X    (ALBF=AL)/COSBF
           GO TO 56
        55 RFLB=(-PX*(R+C)+PZ*AL+AMY+VBF*(ALBF+AL)+VBA*(ALBA-AL))/2./
           X    (ALBF+AL)/COSBF
           RFRB=(-PX*(R+C)+PZ*AL+AMY-VBF*(ALBF+AL)+VBA*(ALBA-AL))/2./
           X    (ALBF+AL)/COSBF
        56 IF(IAB)57,58,58
        57 RALB=0.
           RARB=VBA/COSBA
           GO TO 65
        58 RALB=VBA/COSBA
           RARB=0.
           GO TO 65
C
C    CASE III FORWARD AND AFT SWAYBRACES
C
        59 RZL=0.
           RYL=(PY*E1-AMX)/(R+H-E1)
           RXL=PX
           IF(IFB)60,61,61
        60 RFLB=(-PX*(R+C)+PZ*ALBA+AMY-VBF*(ALBF+ALBA))/2./(ALBF+ALBA)/COSBF
           RFRB=(-PX*(R+C)+PZ*ALBA+AMY+VBF*(ALBF+ALBA))/2./(ALBF+ALBA)/COSBF
           GO TO 62
        61 RFLB=(-PX*(R+C)+PZ*ALBA+AMY+VBF*(ALBF+ALBA))/2./(ALBF+ALBA)/COSBF
           RFRB=(-PX*(R+C)+PZ*ALBA+AMY-VBF*(ALBF+ALBA))/2./(ALBF+ALBA)/COSBF
        62 IF(IAB)63,64,64
        63 RALB=(PX*(R+C)+PZ*ALBF-AMY-VBA*(ALBF+ALBA))/2./(ALBF+ALBA)/COSBA
           RARB=(PX*(R+C)+PZ*ALBF-AMY+VBA*(ALBF+ALBA))/2./(ALBF+ALBA)/COSBA
           GO TO 65
        64 RALB=(PX*(R+C)+PZ*ALBF-AMY+VBA*(ALBF+ALBA))/2./(ALBF+ALBA)/COSBA
           RARB=(PX*(R+C)+PZ*ALBF-AMY-VBA*(ALBF+ALBA))/2./(ALBF+ALBA)/COSBA
C
C    PRINT RESULTS
C
        65 WRITE(6,2004)NCOND,PXA,PX,ANX,PYA,PY,ANY,PZA,PZ,ANZ,AMXA,AMX,
           X             THEDD,AMYA,AMY,PSIDD,AMZA,AMZ
           IF(NLUGS-1)66,67,66
        66 WRITE(6,2005)RZFL,RZAL,RYFL,RYAL,RXL
           GO TO 68
        67 WRITE(6,2006)RZL,RYL,RXL
```


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```
      68 WRITE(6,2007)RFLB,RALB,RFRB,RARB
C
C    READ ANOTHER CONDITION
C
         IF(N-3)3,1,3
      69 WRITE(6,2008)
         STOP

    1000 FORMAT(I1,17A4)
    1001 FORMAT(8F10.0)
    1002 FORMAT(I5,5X,5F10.0)
    2000 FORMAT(45H1LUG AND SWAYBRACE REACTIONS ANALYSIS RESULTS//1X, 17A4)
    2001 FORMAT(/34H GEOMETRY AND CHARACTERISTICS  W =,F7.0,9H LBS  I =,
       X     F11.0,9H LB-IN-SQ/5X,6HALF  =,F6.2,3H IN,6X,7HR    =,
       X     F6.2,3H IN,7X,5HE1  =,F6.2,3H IN/5X,6HALA  =,F6.2,3H IN,
       X     6X,7HC    =,F6.2,3H IN,7X,5HE2  =,F6.2,3H IN/5X,6HALBF =,
       X     F6.2,3H IN,6X,7HH     =,F6.2,3H IN,7X,5HE3   =,F6.2,3H IN/
       X     5X,6HALBA =,F6.2,3H IN,6X,7HBETAF  =,F6.1,4H DEG,6X,
       X     5HSB  =,F6.2/26X,7HBETAA  =,F6.1,4H DEG,6X,5HDEL =,F6.1,
       X     4H DEG/)
    2002 FORMAT(/11X,49H* * * * * * * * * * * * * * * * * * * * * * * * *)
    2003 FORMAT(/34H GEOMETRY AND CHARACTERISTICS  W =,F7.0,9H LBS  I =,
       X     F11.0,9H LB-IN-SQ/5X,6HAL    =,F6.2,3H IN,6X,7HR     =,
       X     F6.2,3H IN,7X,5HE1  =,F6.2,3H IN/26X,7HC     =,F6.2,3H IN,
       X     7X,5HE2  =,F6.2,3H IN/5X,6HALBF  =,F6.2,3H IN,6X,7HH   =,
       X     F6.2,3H IN,7X,5HE3  =,F6.2,3H IN/5X,6HALBA =,F6.2,3H IN,6X,
       X     7HBETAF =,F6.1,4H DEG/26X,7HBETAA =,F6.1,4H DEG,6X,5HDEL =,
       X     F6.1,4H DEG/)
    2004 FORMAT(9H COND NO.,I6/26X,9HPX(AIR) =,F9.0,3H LB,4X,4HPX =,F9.0,
       X     3H LB/3X,7HNX    =,F6.1,10X,9HPY(AIR) =,F9.0,3H LB,4X,
       X     4HPY =,F9.0,3H LB/3X,7HNY    =,F6.1,10X,9HPZ(AIR) =,F9.0,
       X     3H LB,4X,4HPZ =,F9.0,3H LB/3X,7HNZ    =,F6.1, 10X,
       X     9HMX(AIR) =,F9.0,11H IN-LB MX =,F9.0,6H IN-LB/3X,7HTHEDD =,
       X     F6.1,19H /SEC-SQ MY(AIR) =,F9.0,11H IN-LB MY =,F9.0,
       X     6H IN-LB/3X,7HPSIDD =,F6.1,19H /SEC-SQ MZ(AIR) =,F9.0,
       X     11H IN-LB MZ =,F9.0,6H IN-LB/)
    2005 FORMAT(16H  LUG REACTIONS,10X,7HRZFL = ,F7.0,3H LB, 10X,7HRZAL =,
       X     F7.0,3H LB/26X,7HRYFL = ,F7.0,3H LB, 10X,7HRYAL = ,F7.0,
       X     3H LB/41X,6HRXL = ,F7.0,3H LB)
    2006 FORMAT(16H  LUG REACTIONS,25X,5HRXL =,F7.0,3H LB/41X,5HRYL =,
       X      F7.0,3H LB/41X,5HRXL  =,F7.0,3H LB)
    2007 FORMAT(33H  SWAYBRACE REACTIONS   RFLB  = ,F7.0,3H LB,10X,
       X      7HRALB = ,F7.0,3H LB/26X,7HRFRB = ,F7.0,3H LB,10X
       X      7HRARB = ,F7.0,3H LB///)
    2008 FORMAT(1H1)
     END
```


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APPENDIX B

30.2 TDU-34/A Tow Target design data and criteria. The following provides the TDU-34/A Tow Target design data and criteria that were used in establishing the TDU-34/A values for the input definitions in the tow target program.

30.2.1 Airloads Criteria

a. For all tow targets on the RMK-31/A47U/4:

(1) During captive carriage, due to the presence of the reeling machine-launcher, average target angle of attack (α_T) and sideslip angle (β_T) are as follows:

(a) $\alpha_T = 0.4\ (\alpha a/c - 7.1)$, in degrees
(b) $\beta_T = 0.4\ (\beta a/c)$, in degrees

Subscript a/c refers to the conditions appropriate to the carrying aircraft

b. For the TDU-34/A Tow Target:

(1) Overall target derivatives during captive carriage:

(a) $C_D = 0.5$
(b) For $\alpha \leq 0°$, $C_{L\alpha} = 0.1555$ per degree, acting 21 inches aft of the target C.G.
(c) For $\alpha > 0°$, $C_{L\alpha} = 0.1773$ per degree, acting 12.4 inches aft of the target C.G.
(d) $C_{Y\beta} = 0.26595$ per degree, acting 12.4 inches aft of the target C.G.

(2) Local fin parameters during captive carriage.

(a) Derivatives for design of fins and fin attachments will be $C_{L\alpha}$ fin = $C_{Y\beta}$ fin = 0.09641 per degree, for each panel

Derivatives are based on a target cross-sectional area of 0.7854 square feet.

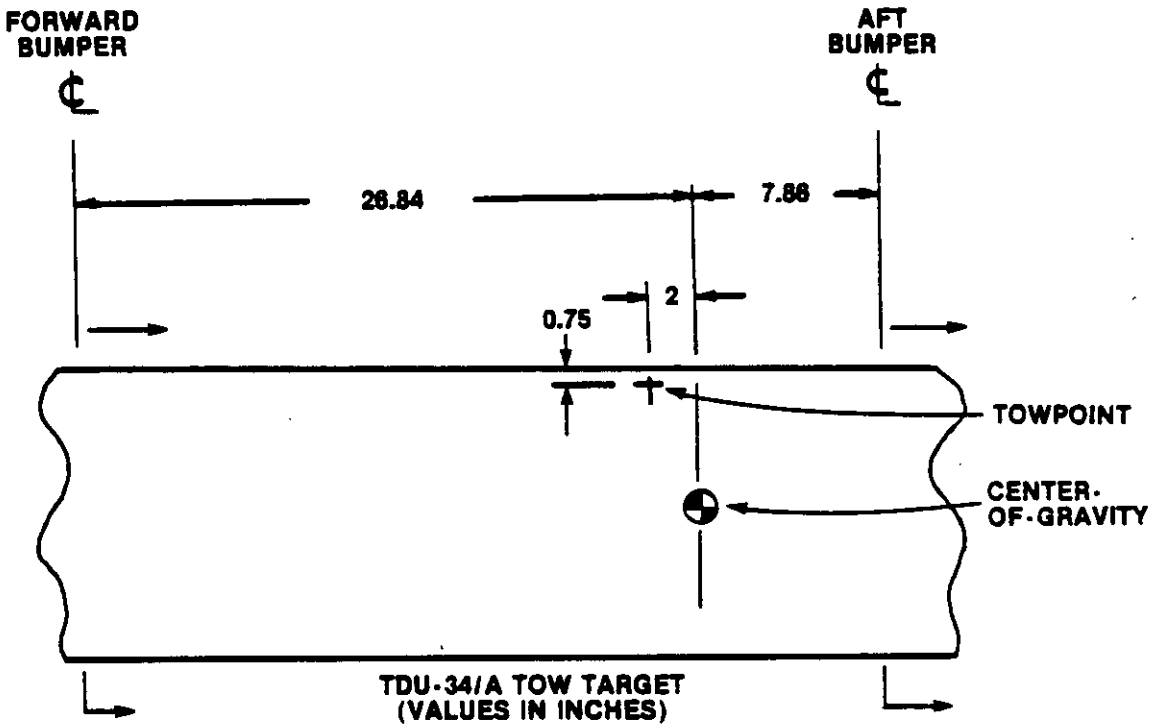
(b) Worst case local angle of attack (α fin) and sideslip angle (β fin) are as follows:

(1) α fin = 16.5° (sin(45°)) = 11.667°, when β fin = 0°
(2) β fin = 16.5° (sin(45°)) = 11.667°, when α fin = 0°

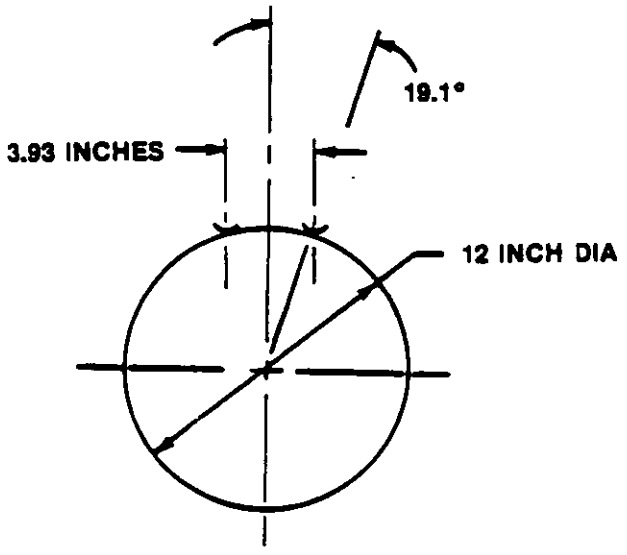
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APPENDIX B

30.2.2 TDU-34/A Tow Target characteristics applicable to the Reeling Machine-Launcher interface with the target.



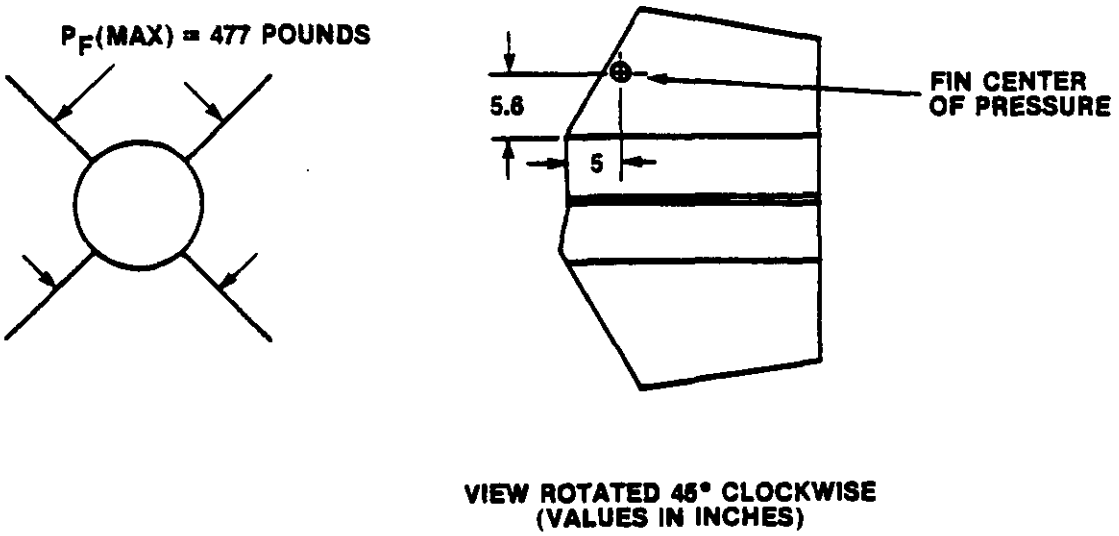
TDU-34/A TOW TARGET
(VALUES IN INCHES)



CROSS SECTION VALUES AT BUMPER ℄

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30.2.3 Target maximum fin load (P_F).



30.2.4 Target physical characteristics.

a. Maximum weight = 200 pounds.

b. Maximum moments of inertia:

(1) I_{xx} = 3555. pounds-inch squared

(2) $I_{yy} = I_{zz}$ = 159120 pounds-inch squared

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30.2.5 Design limit load factors and angles of attack and sideslip for stowed tow target on RMK-31/A47U-4 Reeling Machine Launcher.

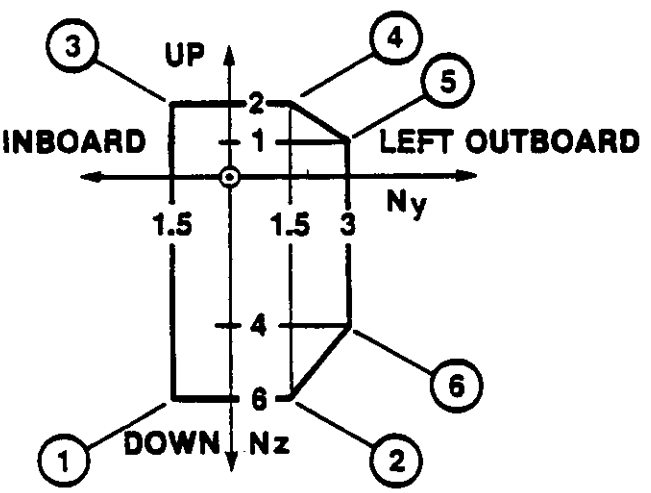
a. The following notes are typical to the following stowed flight, catapult and arrested landing design loads.

(1) $\ddot{\theta}$ = PITCH ACCELERATION, RAD/SEC²

(2) $\ddot{\psi}$ = YAW ACCELERATION, RAD/SEC²

(3) TARGET AXES ROTATED 7.1 DEGREES NOSE DOWN RELATIVE TO AXES SHOWN

30.2.5.1 Stowed flight load factors



N_X	± 1.5
$\ddot{\theta}$	± 4
$\ddot{\psi}$	± 2

ANGLES OF ATTACK & SIDESLIP 400 KEAS, q = 540 LBS/FT²		
POINT	α TARGET (°)	β TARGET (°)
(1) & (2)	-2.9 TO 5.1	± 1.4
(3) & (4)	-7.6 TO -2.9	± 1.4
(5)	-7.9 TO -1.0	± 6.6
(6)	-2.9 TO 5.4	± 6.6

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30.2.5.1 (cont'd)

STOWED TARGET FLIGHT AIR LOADS

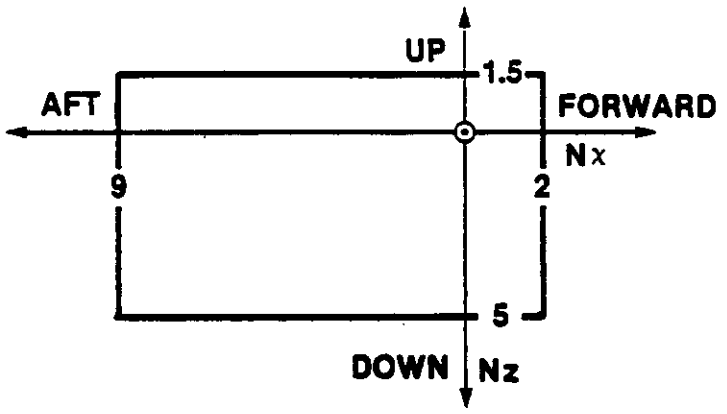
POINT	AP_x	α_T	AP_z	AM_y	β_T	AP_y	AM_z
1 & 2	212	– 2.9	– 192	4,015	± 1.4	± 158	± 1,958
	212	5.1	384	– 4,753	± 1.4	± 158	± 1,958
3 & 4	212	– 7.8	– 501	10,521	± 1.4	± 158	± 1,958
	212	– 2.9	– 192	4,015	± 1.4	± 158	± 1,958
5	212	– 7.9	– 521	10,936	± 6.6	± 745	± 9,227
	212	– 1.0	– 66	1,385	± 6.6	± 745	± 9,227
6	212	– 2.9	– 192	4,015	± 6.6	± 745	± 9,227
	212	5.4	406	– 5,033	± 6.6	± 745	± 9,227

NOTES:

1. AP_x = AIRLOAD, X AXIS, LBS, AFT POSITIVE
2. α_T = TARGET ANGLE OF ATTACK, DEGREES, NOSE UP POSITIVE
3. AP_z = AIRLOAD, Z AXIS, LBS, UP POSITIVE
4. AMy = PITCHING MOMENT DUE AP_z, LB-IN, NOSE UP POSITIVE
5. β_T = TARGET ANGLE OF SIDESLIP, DEGREES, NOSE LEFT POSITIVE
6. AP_y = AIRLOAD, Y AXIS, LBS, OUTBOARD AND LEFT POSITIVE
7. AM_z = YAWING MOMENT DUE AP_y, LB-IN, NOSE LEFT POSITIVE

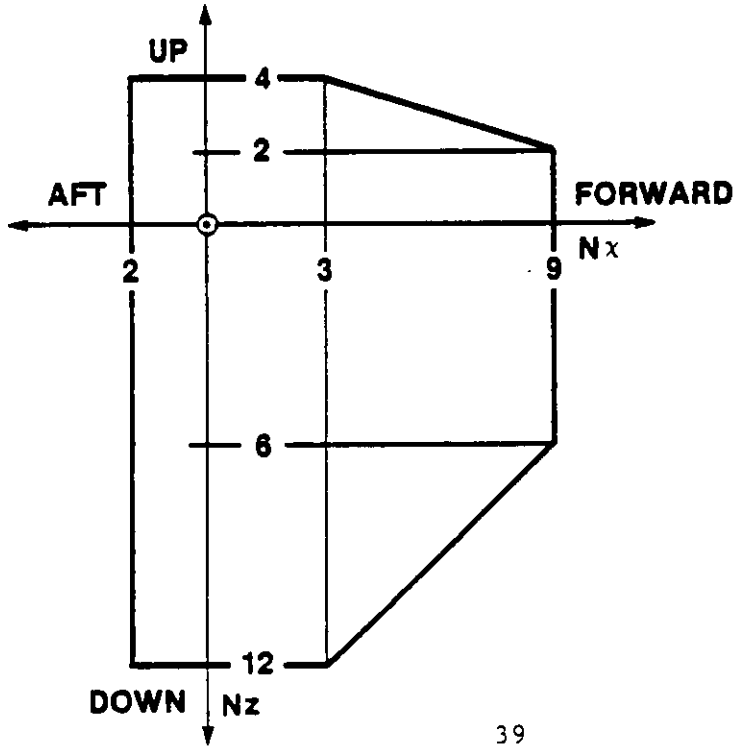
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30.2.5.2 Stowed catapult load factors.



Ny	± 2.5
$\ddot{\theta}$	± 15
$\ddot{\psi}$	± 4

30.2.5.3 Stowed arrested landing load factors.



Ny	± 5.0
$\ddot{\theta}$	± 25
$\ddot{\psi}$	± 6

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APPENDIX B

30.2.6 Input parameters for the TDU-34/A tow target store computer program.

a. Card 1 - Title Card

Column 1 - Number of Suspension Lugs, (1)
Columns 2-69 - Title of Computer Run, (TDU-34/A)

b. Card 2 - Launcher Geometry

Columns 1-10 - Angle between the launcher centerline and the vertical plane in a fore-and-aft view, in degrees, (0.0)
Columns 11-20 - Distance from the target center of gravity to the center of the tow-bolt, in inches, positive aft of c.g., (-2.0)
Columns 21-30 - [Only used for multiple lug configurations], (0.0)
Columns 31-40 - Distance from the target c.g. to the forward bumper, in inches, positive forward of c.g., (26.84)
Columns 41-50 - Distance from the target c.g. to the aft bumper, in inches, positive aft of the c.g., (7.86)
Columns 51-60 - Angle between the launcher centerline and the line of action of the forward bumpers, in degrees, (19.1)
Columns 61-70 - Angle between the launcher centerline and the line of action of the aft bumpers, in degrees, (19.1)
Columns 71-80 - [Only used for multiple lug configurations], (0.0)

c. Card 3 - Store Characteristics

Columns 1-10 - Weight of the target, in pounds, (200.)
Columns 11-20 - Pitch and yaw mass moment of inertia, in pound-inches squared, (159120.)
Columns 21-30 - Radius of the target fuselage, in inches, (6.0)
Columns 31-40 - Distance from the upper surface of the target to the center of the tow-bolt, positive above surface, in inches, (-0.75)
Columns 41-50 - Same as columns 31-40, (-0.75)
Columns 51-60 - Vertical distance from the target geometric center to the point of intersection of the bumper lines of action, positive above the geometric center, in inches, (0.0)
Columns 61-70 - Vertical distance from the target geometric center to the center of gravity, positive below the geometric center, in inches, (0.0)
Columns 71-80 - Lateral distance from the target geometric center to the center of gravity, positive to the right of the geometric center looking forward, in inches, (0.0)

d. Card 4 - Flight Conditions

Columns 1-5 - Flight Condition Number, (An integer, right justified)
Columns 6-10 - BLANK
Columns 11-20 - Axial design limit load factor, (Nx, decimal number)
Columns 21-30 - Lateral design limit load factor, (Ny, decimal number)

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Columns 31-40 - Vertical design limit load factor, (Nz, decimal number)
Columns 41-50 - Pitch acceleration, in radians per second squared, ($\ddot{\theta}$, decimal number)
Columns 51-60 - Yaw acceleration, in radians per second squared, ($\ddot{\psi}$, decimal number)
Columns 61-80 - BLANK

e. Card 5 - Airloads

Columns 1-10 - Axial air load, positive aft, in pounds, (APx, decimal number)
Columns 11-20 - Lateral air load, positive left looking forward, in pounds, (APy, decimal number)
Columns 21-30 - Vertical air load, positive up, in pounds, (APz, decimal number)
Columns 31-40 - Rolling moment induced by air loads, positive counterclockwise looking forward, in inch-pounds, (0.0)
Columns 41-50 - Pitching moment induced by air loads, positive nose-up, in inch-pounds, (AMy, decimal number)
Columns 51-60 - Yawing moment induced by air loads, positive nose left, in inch-pounds, (AMz, decimal number)
Columns 61-80 - BLANK

f. Card 6 - Run Terminator Card

Columns 1-80 - BLANK

NOTE: Cards 4 and 5 are repeated in sets until all combinations of flight condition and airloads have been run. The final set is followed by card 6.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-85230(AS)	TARGET, TOW, AERIAL, TDU-34/A

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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